

ME-2412

**Engineering Drawing
Lab**

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Text book:

- Mechanical Engineering Drawing
-by Dr. Md. Quamrul Islam
-

Reference books :

- Fundamentals of Engineering Drawing
-by French & Vierck.
- Metric Drafting
– by Paul Wallah.
- Drafting Technology and Practice
– by William P. Spence

Graphics Language

Effectiveness of Graphics Language

1. Try to write a description of this object.
 2. Test your written description by having someone attempt to make a sketch from your description.
-



You can easily understand that ...

The word languages are inadequate for describing the **size**, **shape** and **features** completely as well as concisely.

Composition of Graphic Language

Graphic language in “**engineering application**” use **Lines** to represent the **surfaces**, **edges** and **Contours** of objects.

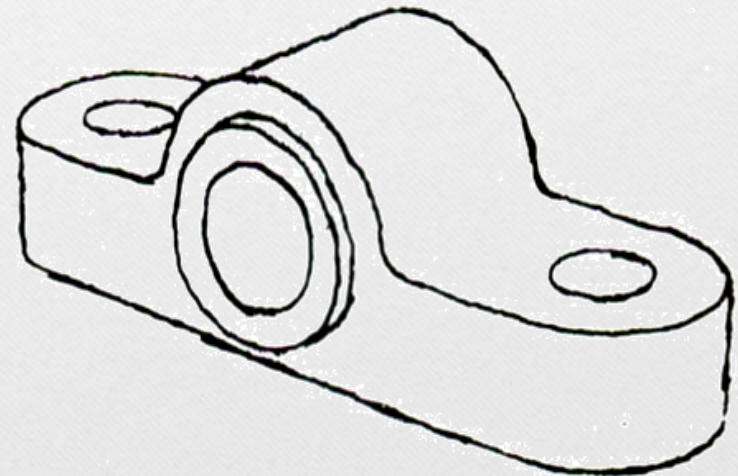
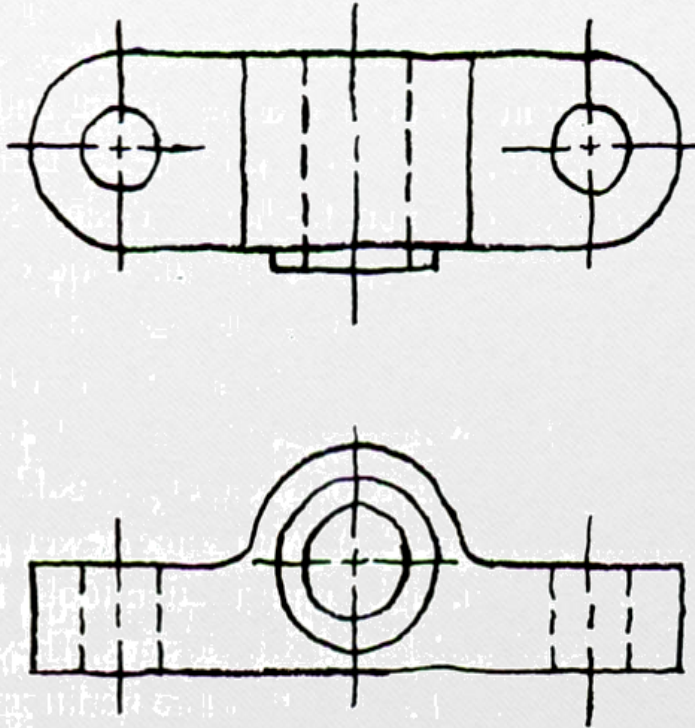
The language is known as “**drawing**” or “**drafting**” .

A drawing can be done using **freehand**, **instruments** or **computer** methods.

Freehand drawing

The lines are sketched without using instruments other than pencils and erasers.

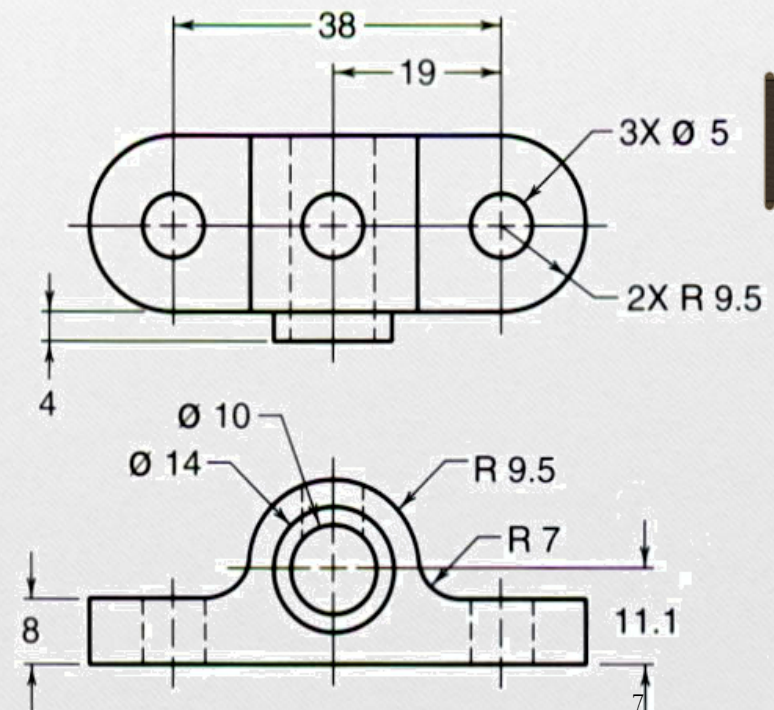
Example



Instrument drawing

Instruments are used to draw straight lines, circles, and curves concisely and accurately. Thus, the drawings are usually made to scale.

Example



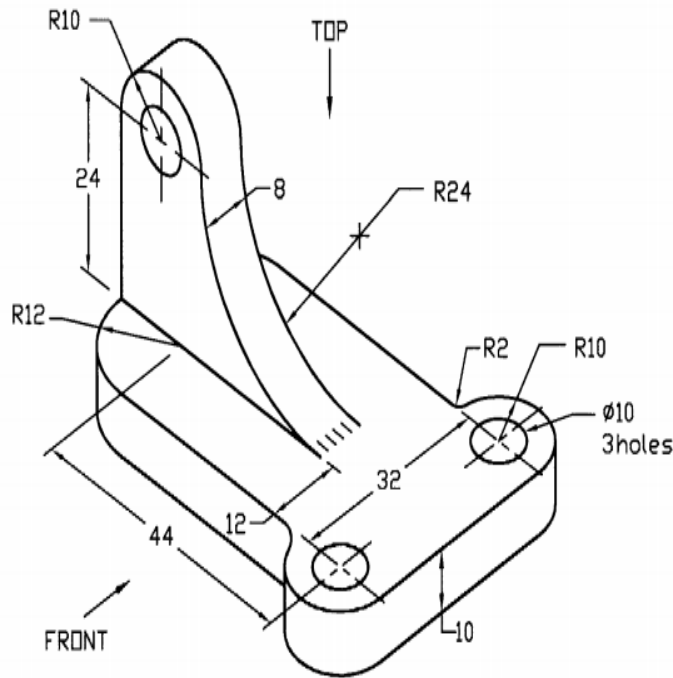
Computer drawing

The drawings are usually made by commercial software such as AutoCAD, solid works etc.

Example



Engineering Drawing

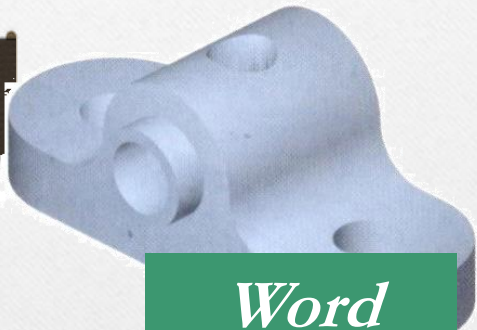


Suppose you want the block shown in the figure for certain project/thesis. Manufacturer will want to know the dimensions, hole no. etc. for building this object.

The end goal of an engineering drawing is to convey all the required information that will allow a manufacturer to produce that component.

Elements of Engineering Drawing

Engineering drawing are made up of **graphics language** and **word language**.



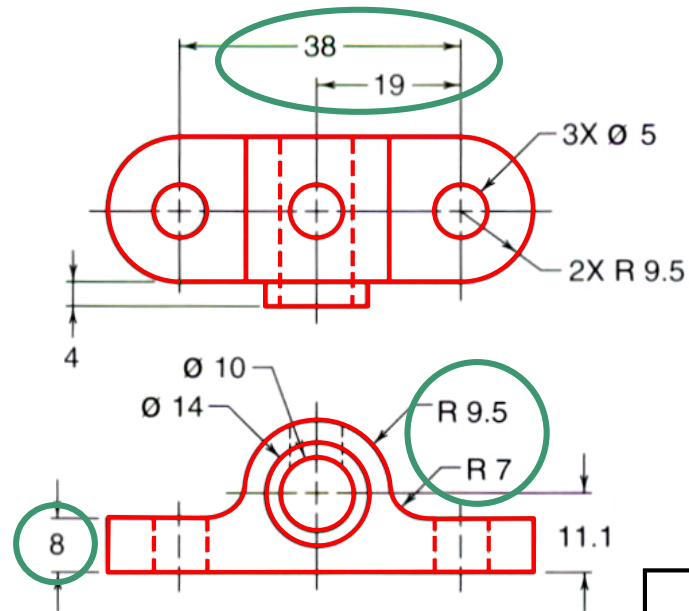
**Word
language**

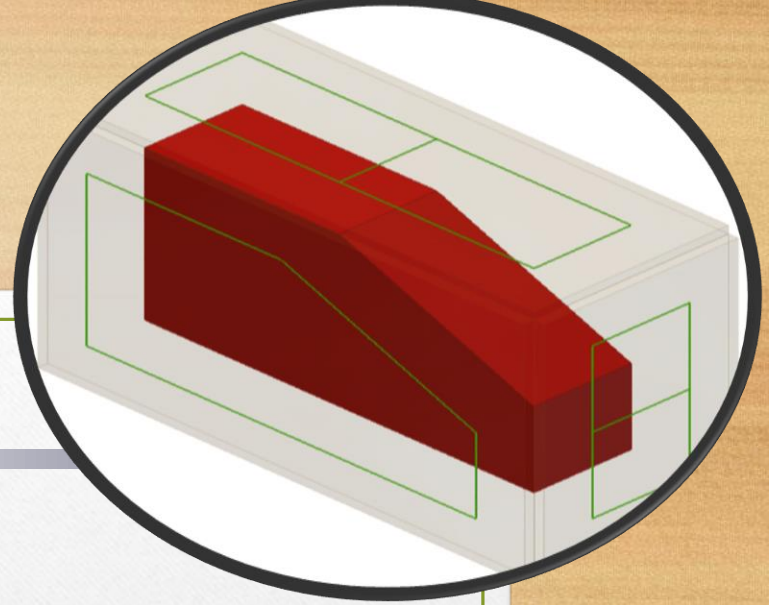
Describe size,
location and
specification of the
object.



Graphics

Describe a shape
(mainly).





PROJECTION METHOD

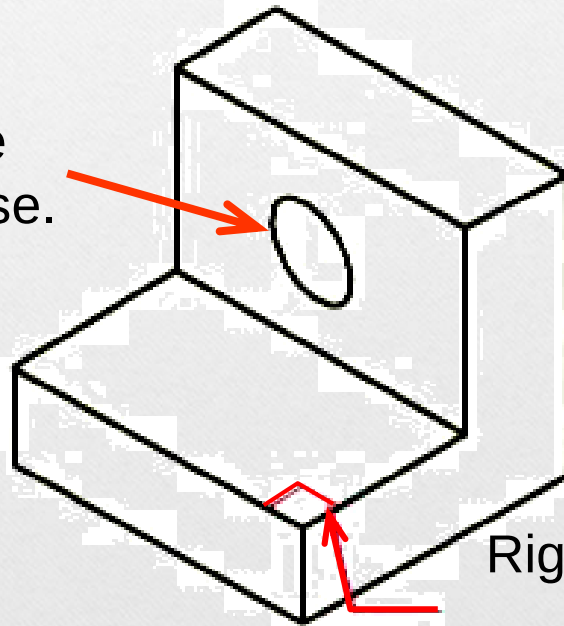
Isometric Drawing

Advantage Easy to understand

Disadvantage Shape and angle distortion

Example Distortions of shape and size in isometric drawing

Circular hole
becomes ellipse.



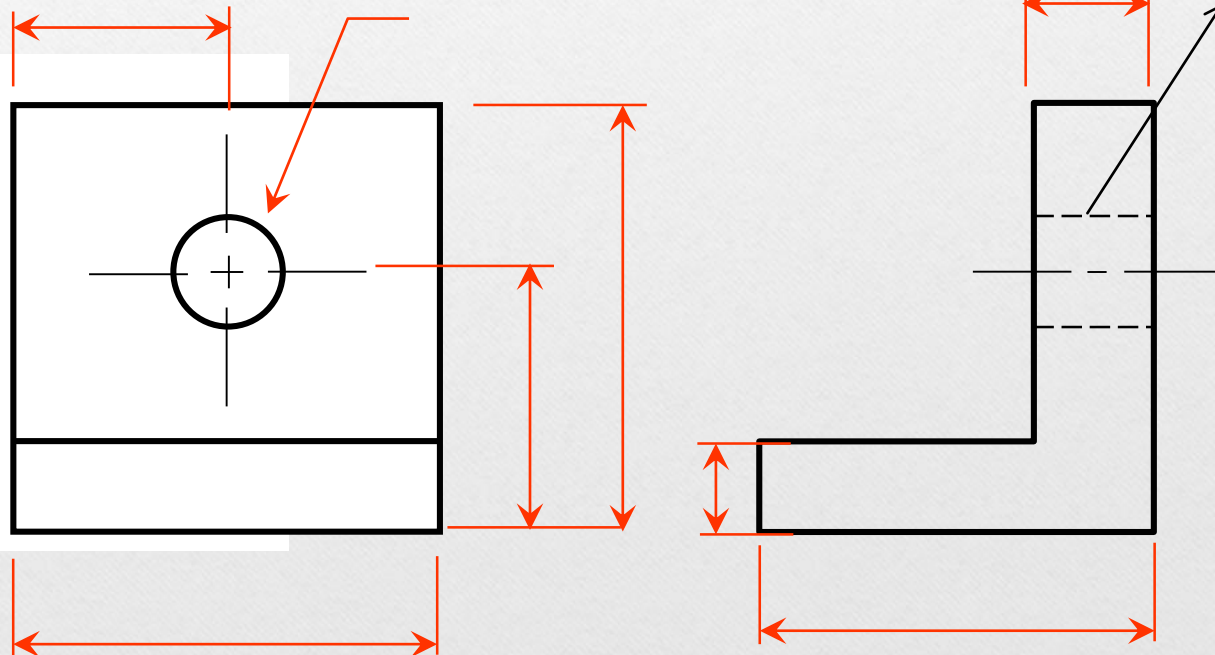
Right angle becomes obtuse
angle.

Multi-view Drawing

Advantage It represents accurate **shape and size**.

Disadvantage Require practice in writing and reading.

➤ Hidden lines show details that are not seen in all views



PROJECTION THEORY

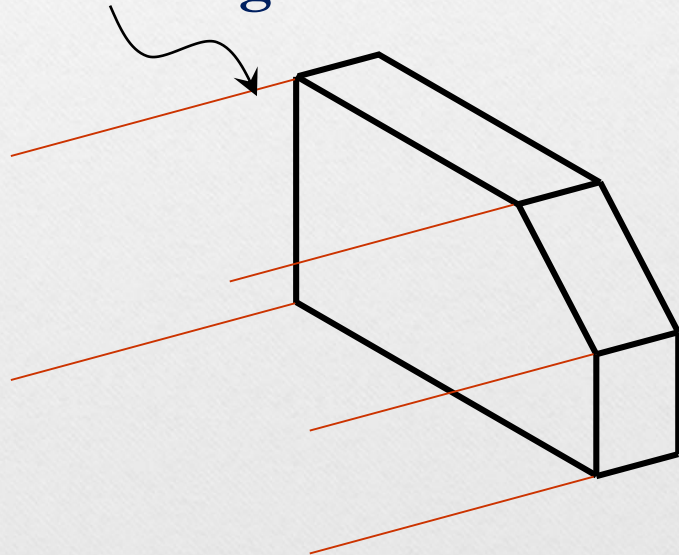
- The projection theory is used to graphically represent 3-D objects on 2-D media (paper, computer screen).
- The projection theory is based on two variables:
 - 1) Line of sight
 - 2) Plane of projection (image plane or picture plane)

Line of sight is an imaginary ray of light between an observer's eye and an object.

There are 2 types of LOS : Parallel and Perspective

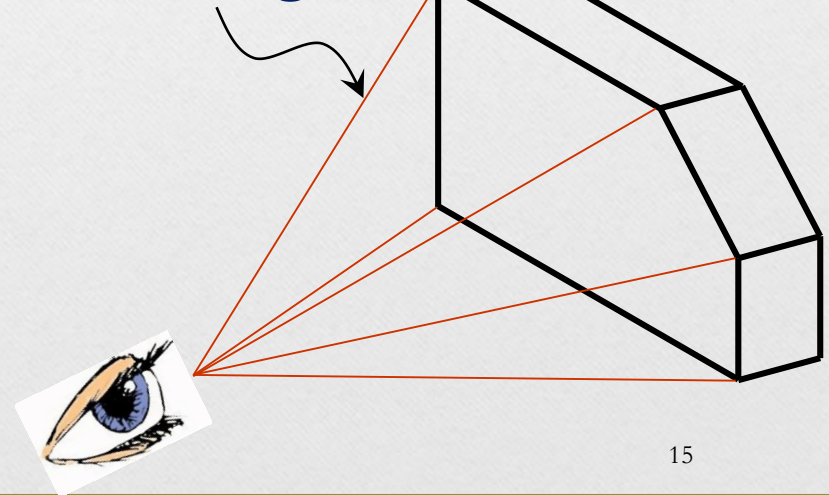
Parallel projection

Line of sight



Perspective projection

Line of sight

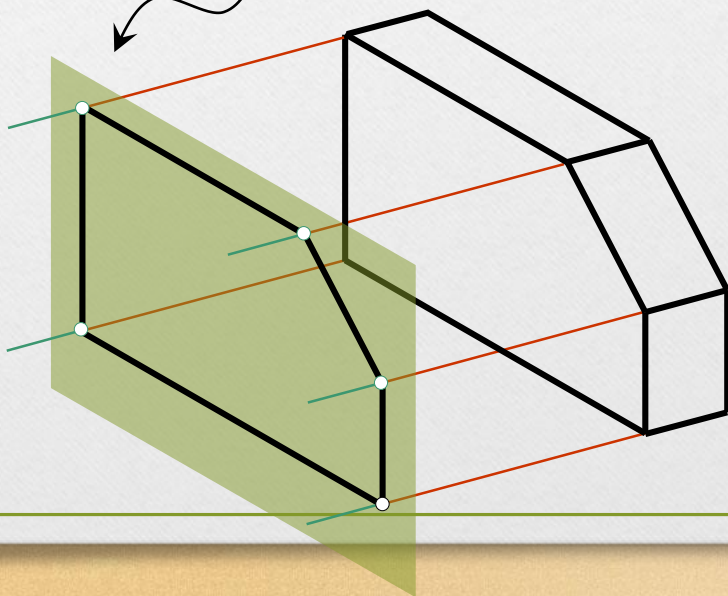


Plane of projection is an imaginary flat plane which the image is created.

The image is produced by connecting the points where the LOS pierce the projection plane.

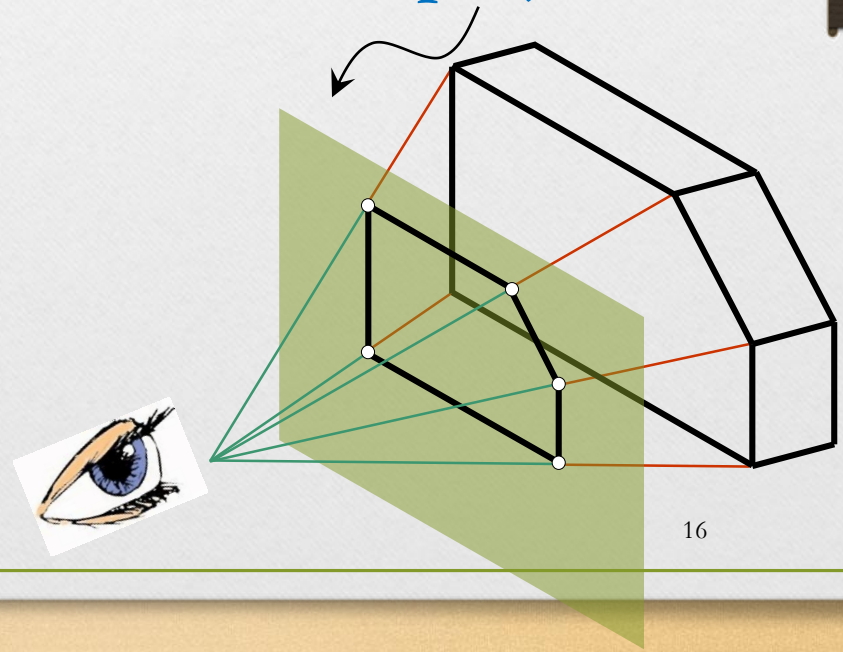
Parallel projection

Plane of projection



Perspective projection

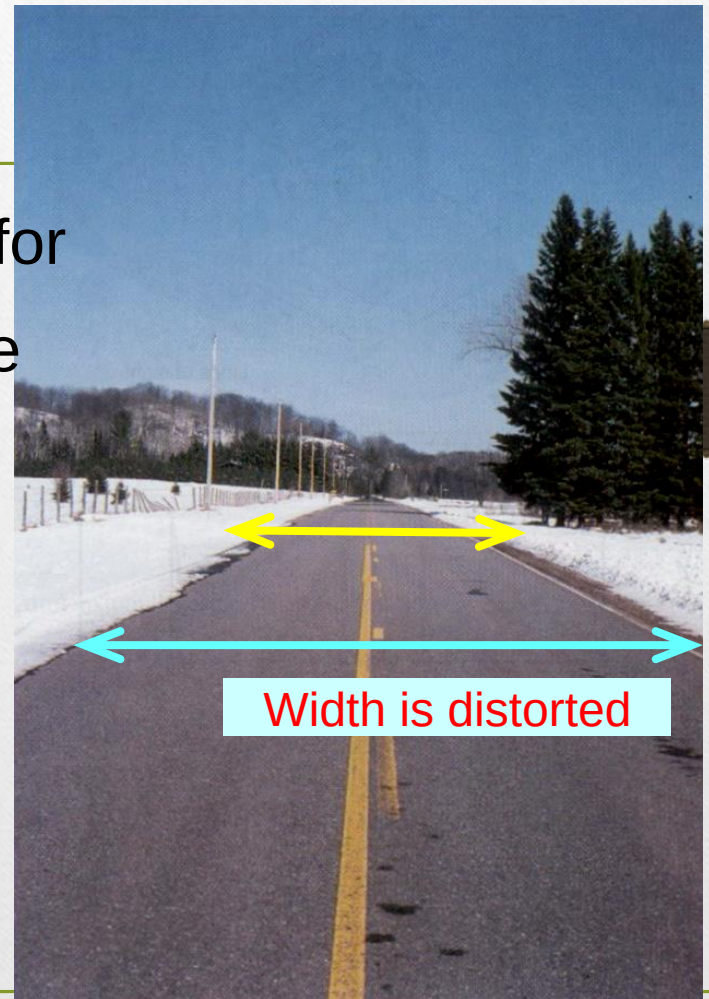
Plane of projection



Disadvantage of Perspective Projection

❖ Perspective projection is *not*
used by engineer for
manufacturing of parts, because

- 1) It is difficult to create.
- 2) It does not reveal exact
shape and size.



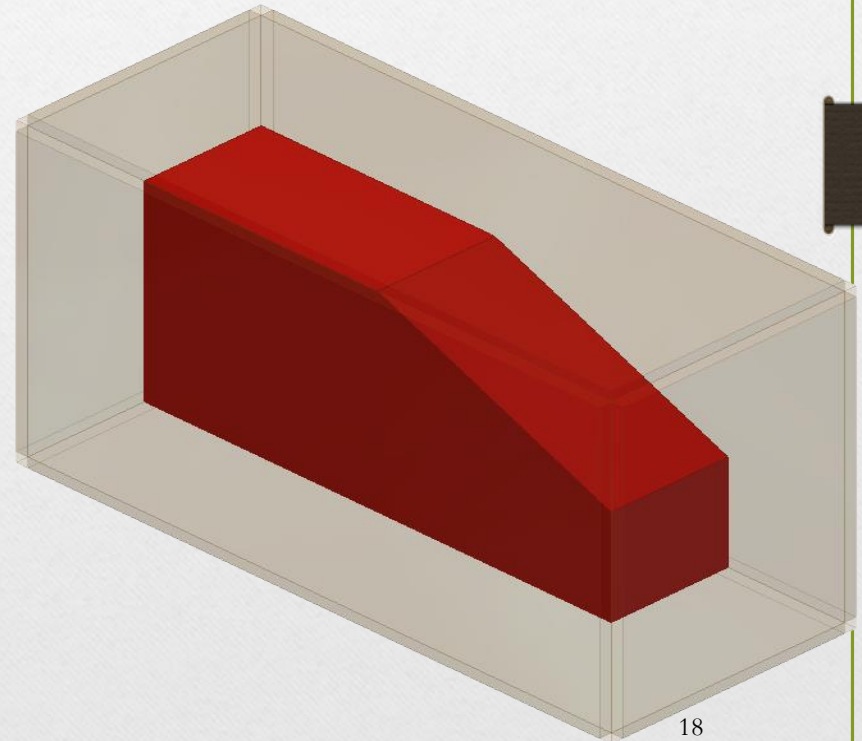
Orthographic Projection

MEANING

Orthographic projection is a parallel projection technique in which the parallel lines of sight are *perpendicular* to the projection plane.

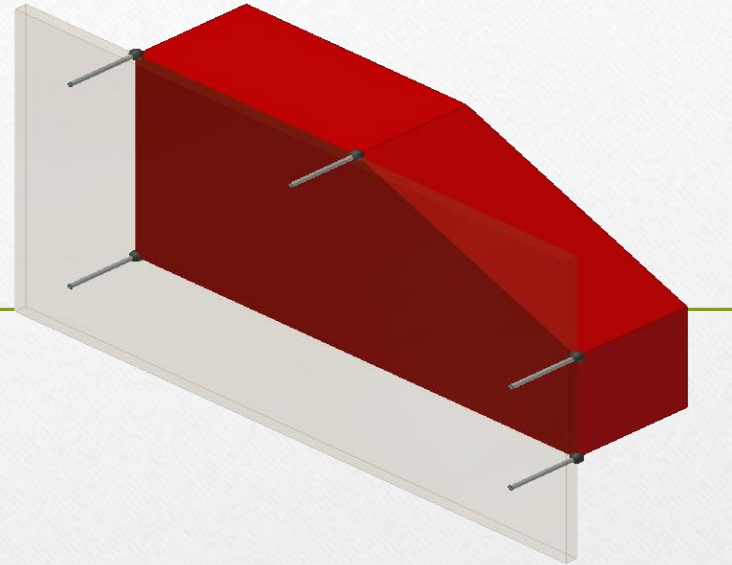
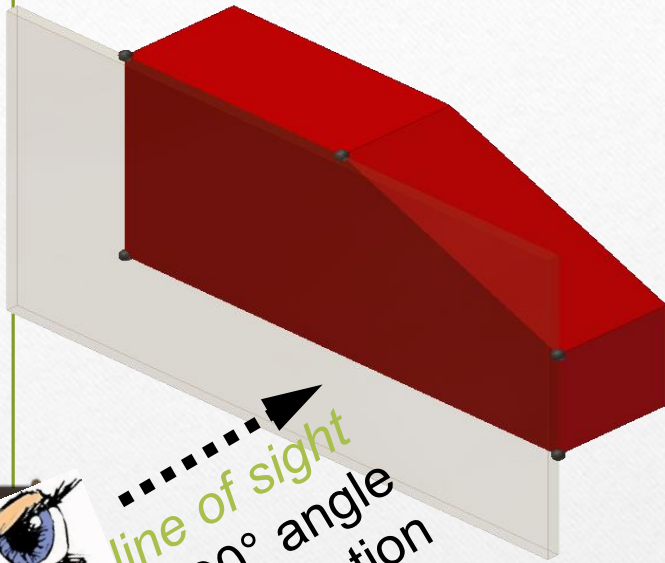
The best way to understand *orthographic projection* is to imagine an object contained inside a glass box.

Each wall represents a *projection plane* onto which a two-dimensional object view will be created.





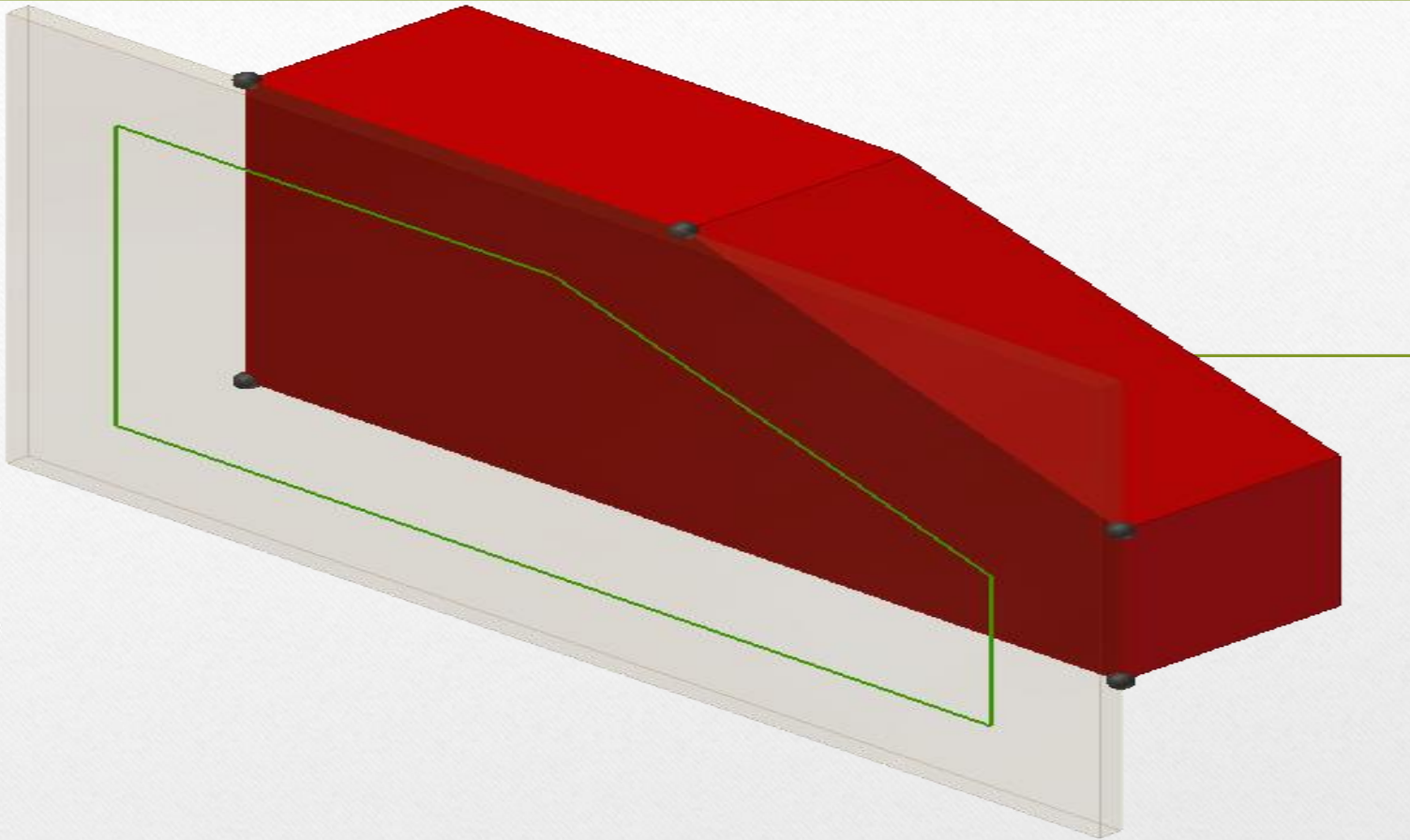
line of sight
at 90° angle
to projection
plane



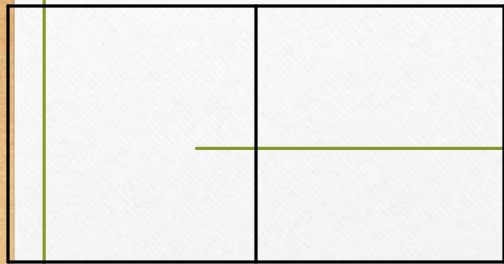
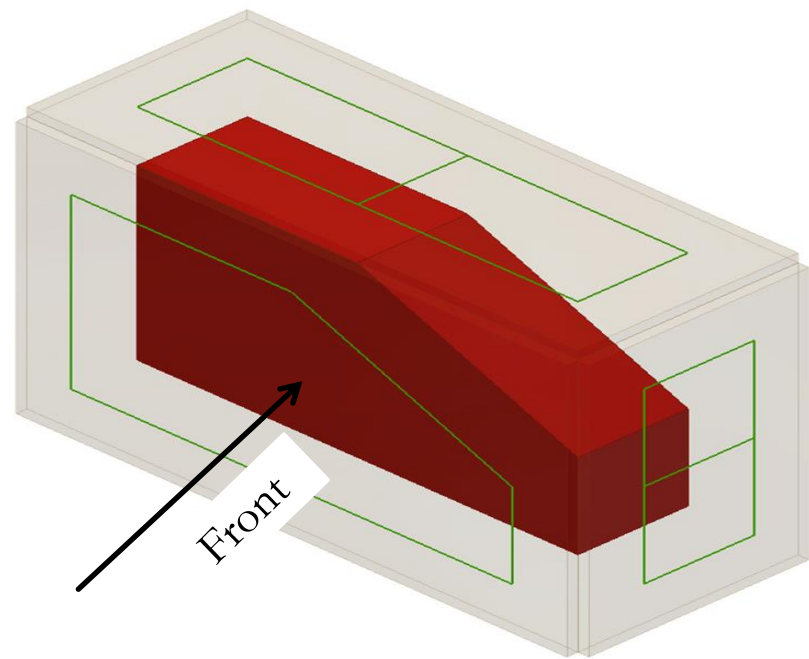
Start by focusing only on the front
projection plane.

A person standing in front of the
object would see only the five
corners identified in black.

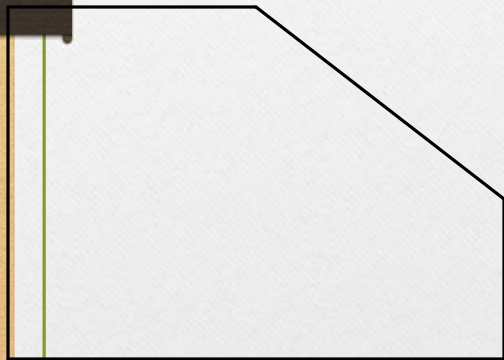
Projection lines are used to project
each corner outward until they
reach the *projection plane*.



The *visible edges* of the object are then identified on the *projection plane* by connecting the projected corners with *object lines*.



TOP VIEW



FRONT
VIEW



RIGHT HAND
SIDE VIEW

Notes:

1. **Front** view is always central
2. **Top** view goes above
3. **R.Side** view goes to the right
4. L.Side view goes to the left.

Orthographic Projection

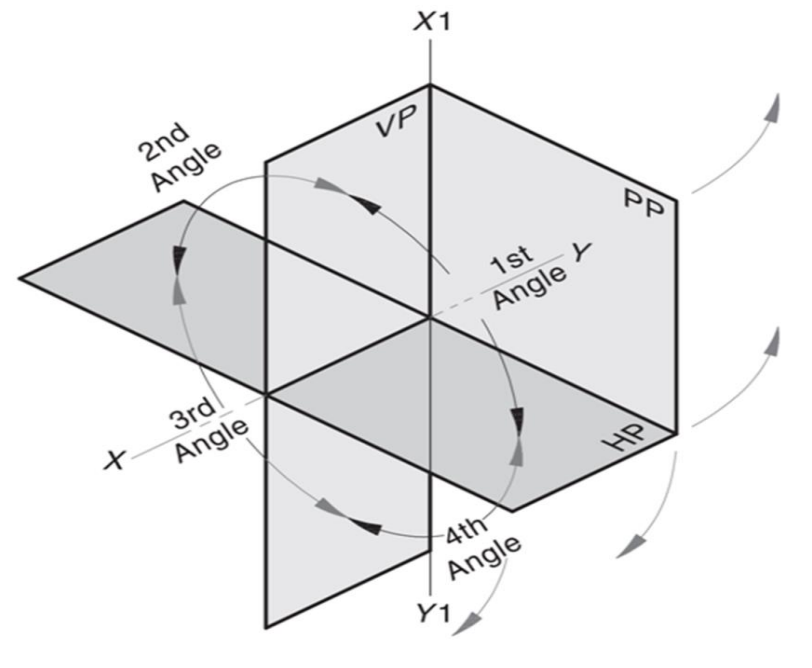
- **PROJECTION SYSTEMS**

1. **First** angle system

- European country

2. **Third** angle system

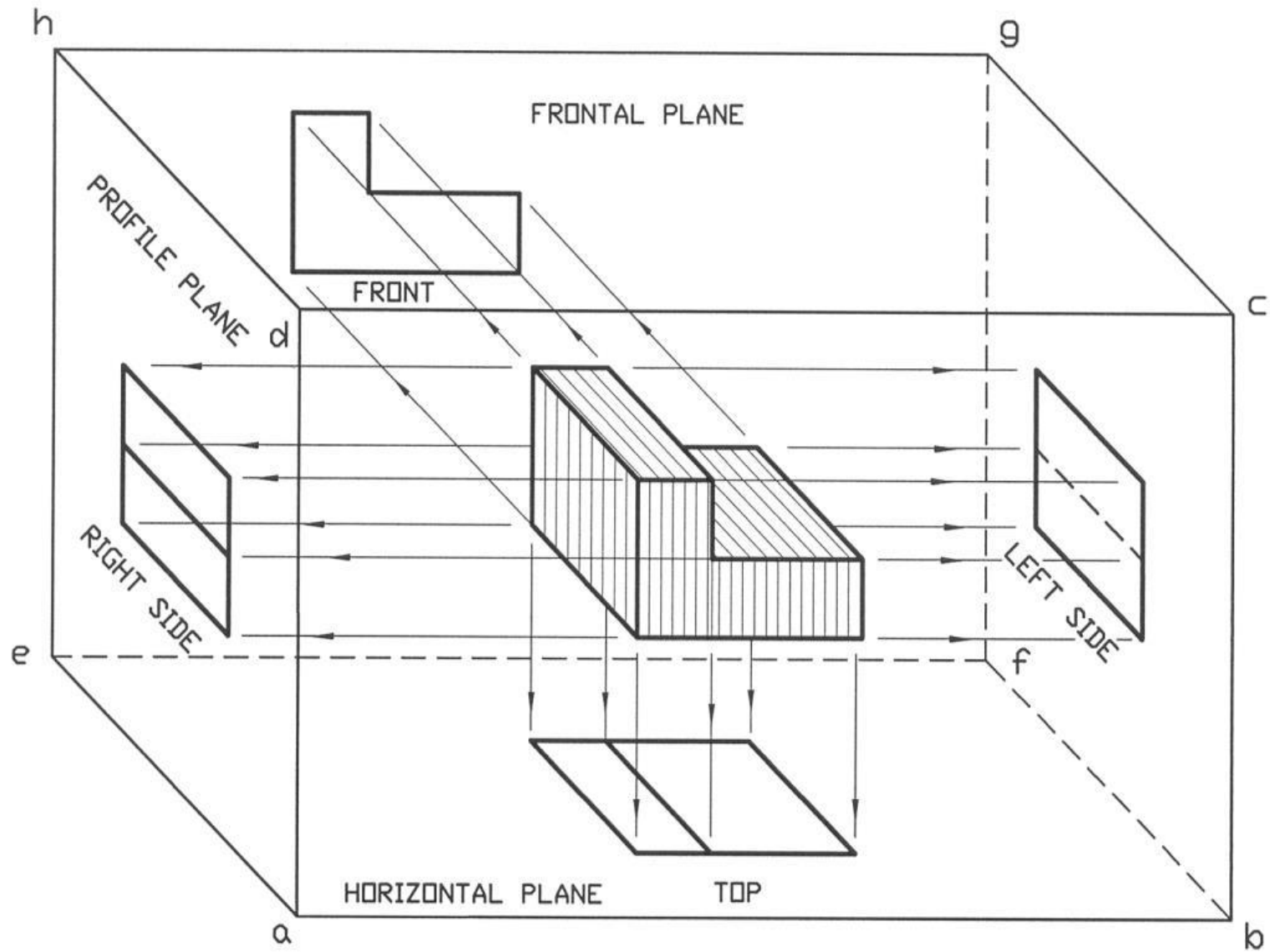
- Canada, USA, Japan, Thailand



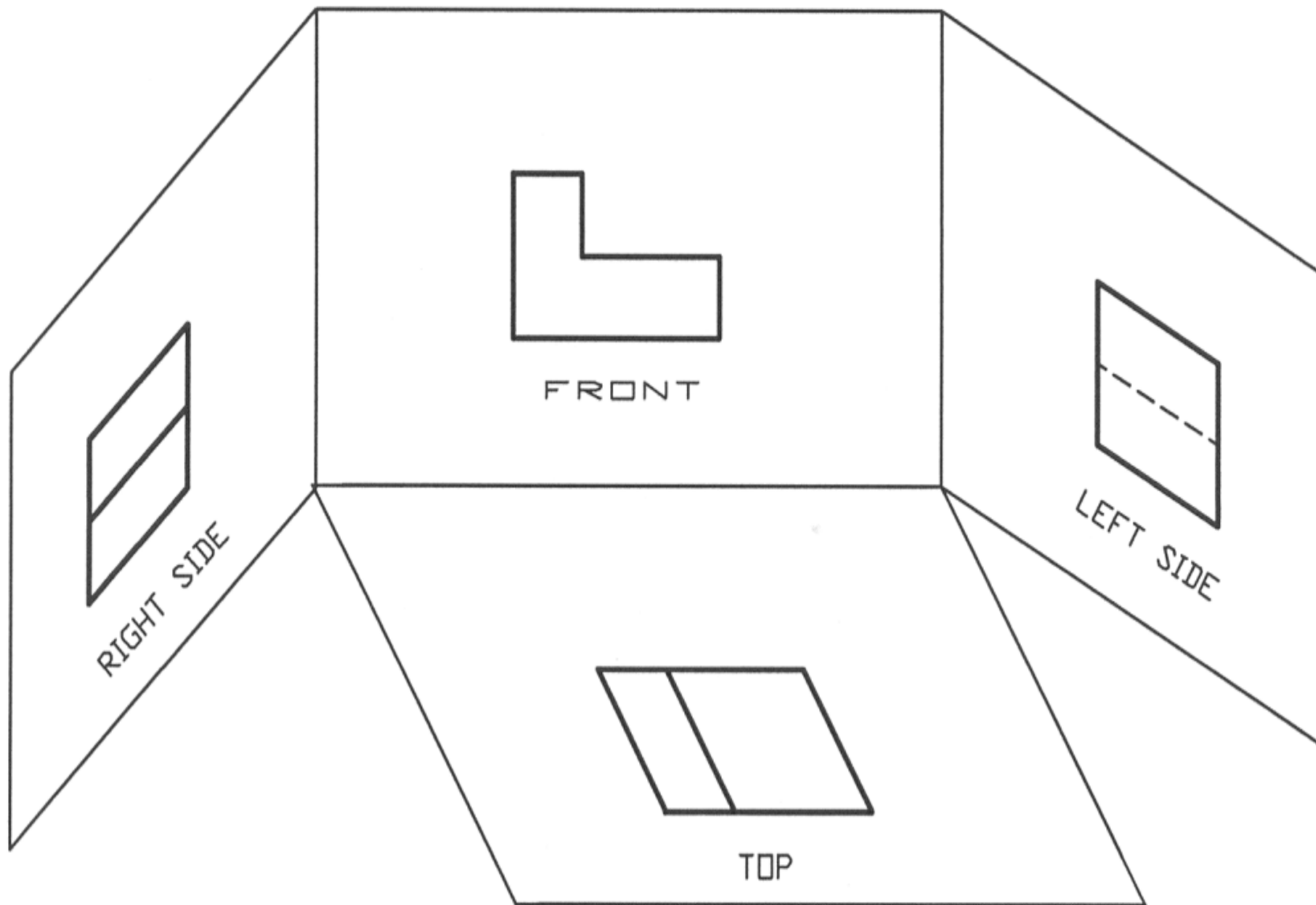
HP- Horizontal Plane;

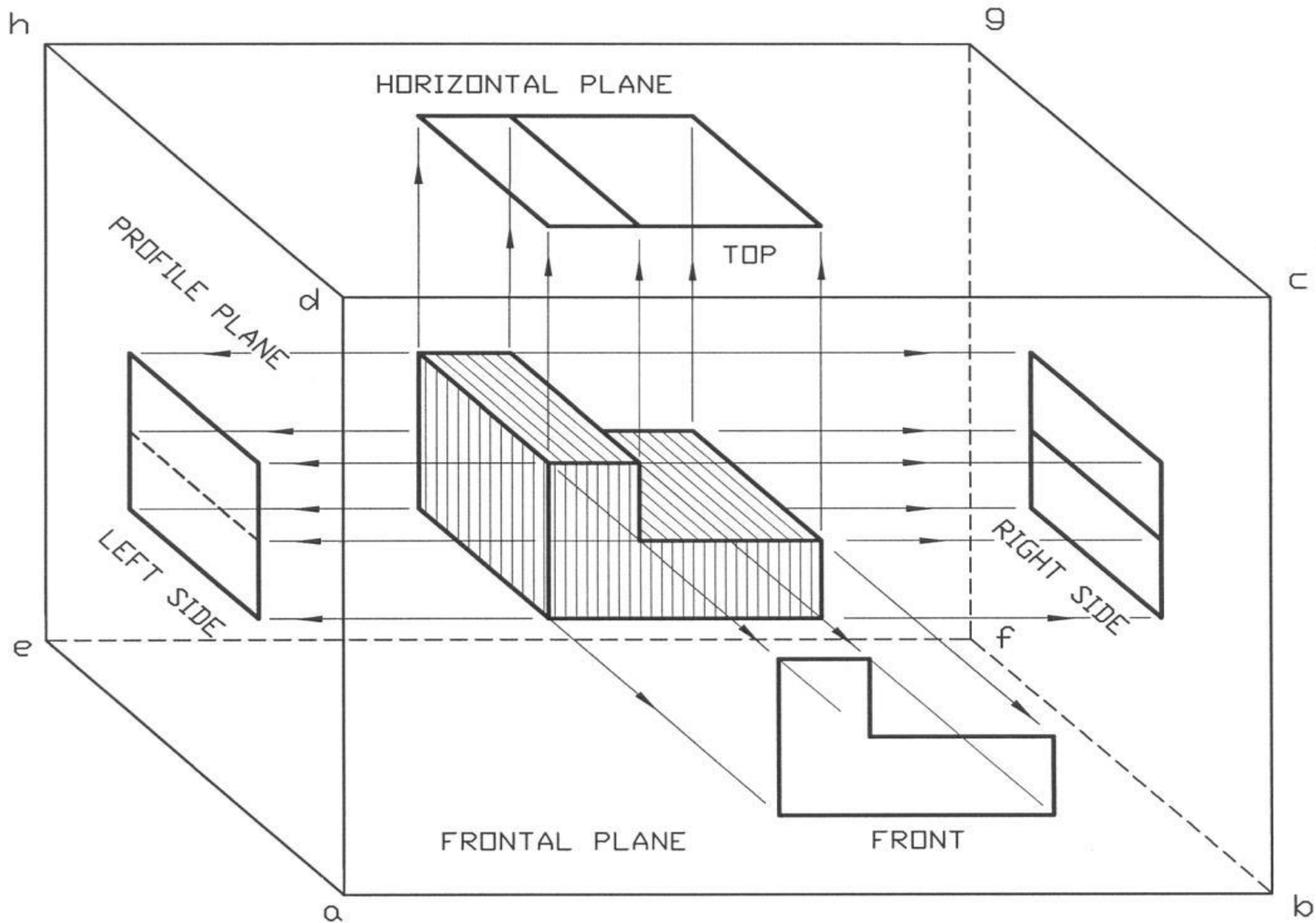
VP-Vertical Plane;

PP-Profile plane

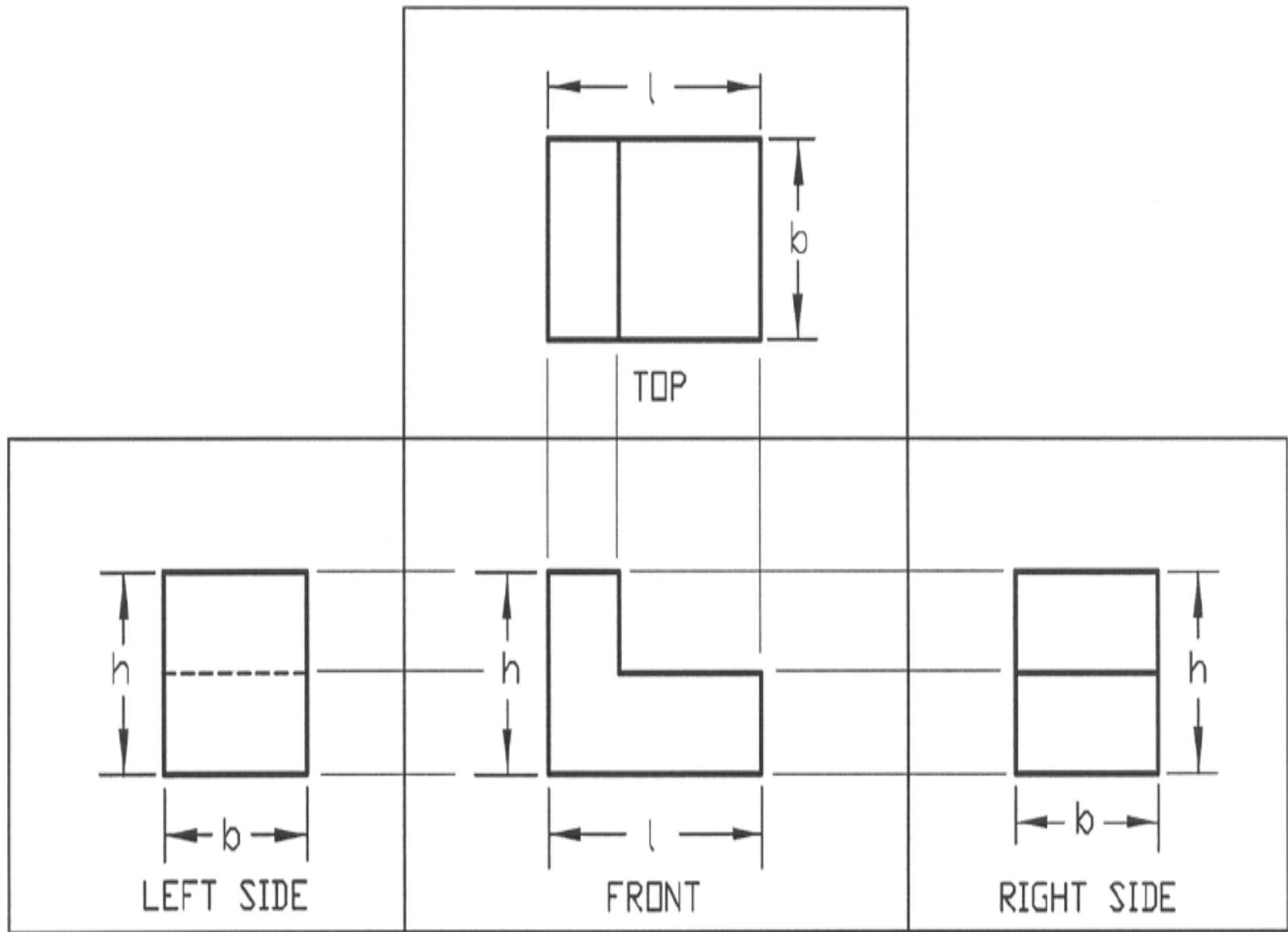


First Angle projection





Third Angle Projection. Which one is Easier?

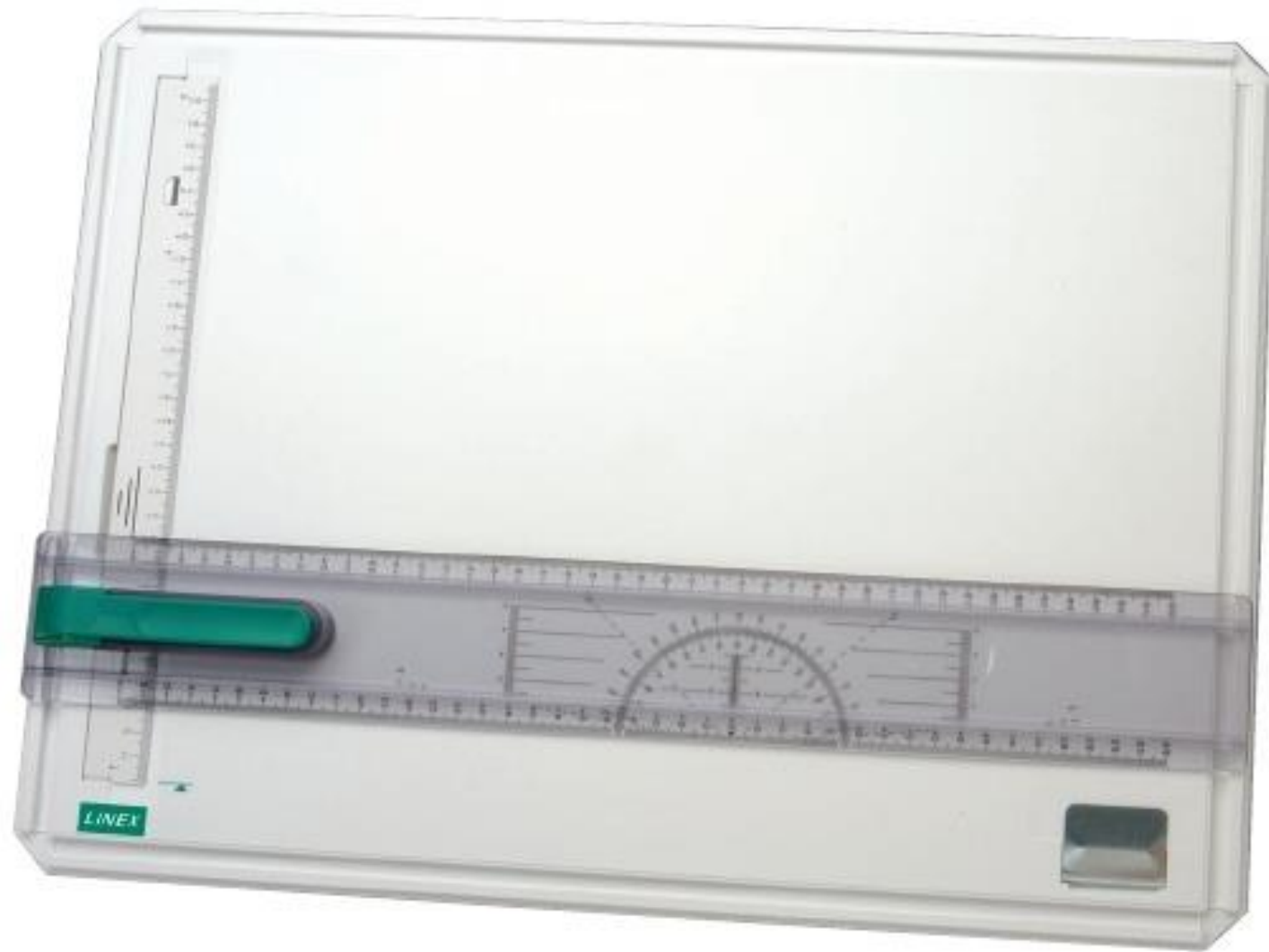


Relative Positions of Views on Sides of Unfolding Box in Third Angle Projection ₂₆

Traditional Drawing Instruments...

- Drawing board/table.
- Drawing sheet/paper.
- Drafting tape.
- Pencils.
- Eraser.
- Sharpener.
- T-square.
- Set-squares/triangles.
- Scales.
- Compass and divider.

Drawing board

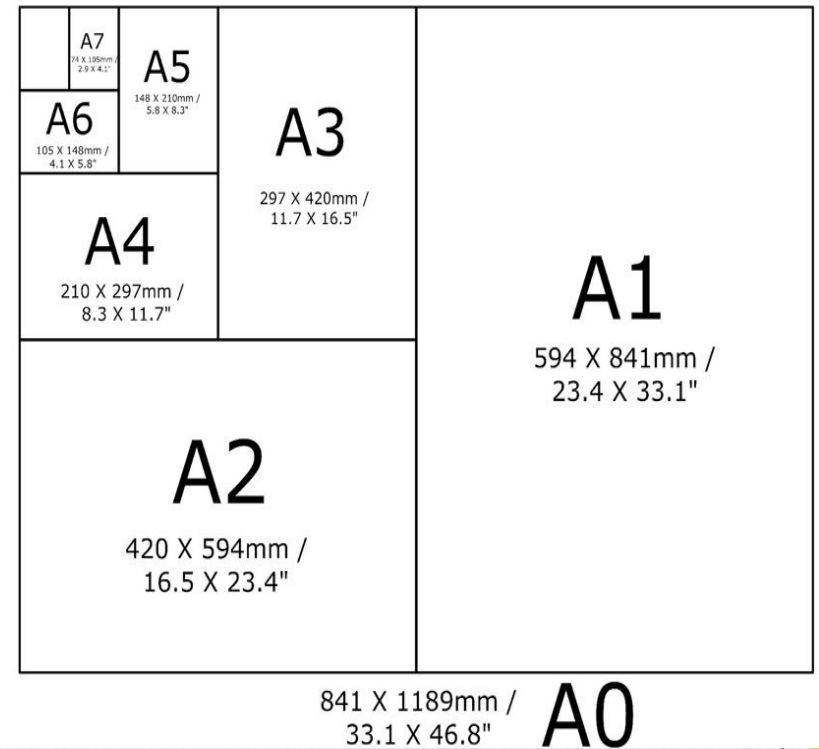


Drawing table



Paper Size

Paper Size	Border Size (bmm * 1mm)	Overall Paper Size (b ₀ mm * l ₀ mm)
A0	821*1159	841*1189
A1	574*811	594*841
A2	400*564	420*594
A3	277*390	297*420
A4	190*267	210*297



Drawing sheet/paper

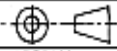
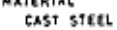
- 216 X 280 mm
- 280 X 382 mm
- 382 X 560 mm
- 585 X 726 mm



DO NOT SCALE

REVISIONS			
SYM	DESCRIPTION	DATE	APPD.
A	N12 WAS 1/2 WHIT.	14-12-78	A.W.B.

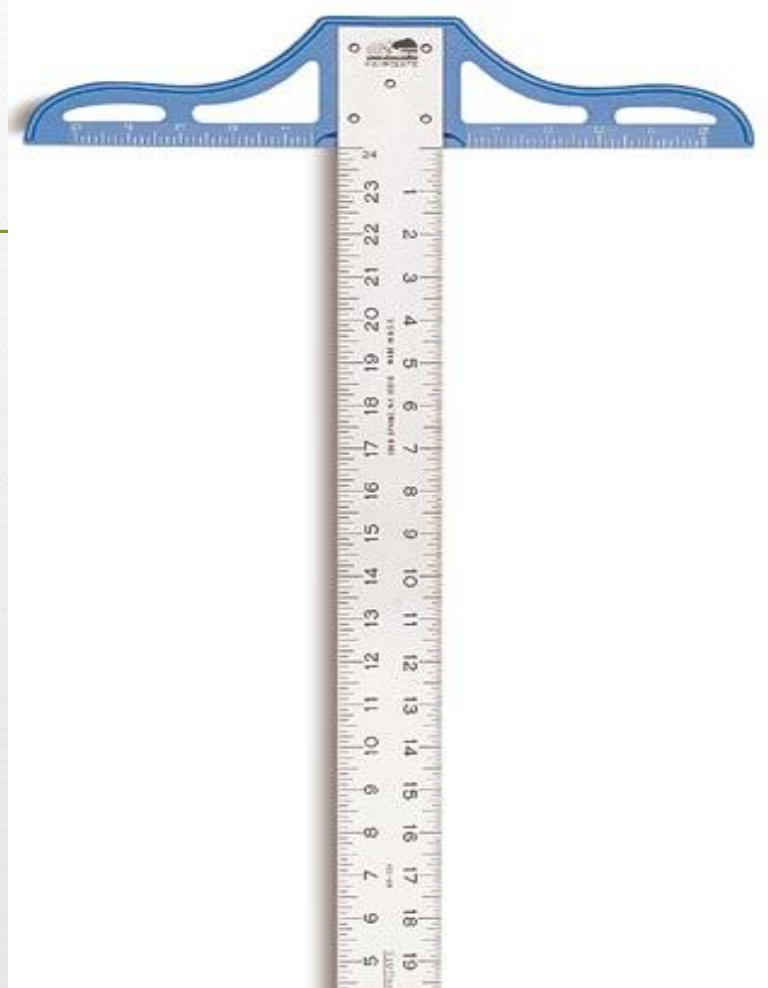
4			
3			
2			
1	A7325	VALVE BODY	1
TEX	DRG or PART No.	DESCRIPTION	QTY

UNLESS OTHERWISE STATED ALL DIMENSIONS IN MILLIMETRES. TOLERANCES LINEAR: ANGULAR:			DRN 1:1:78 JKL	(NAME OF FIRM)
			CKD 2:1:78 MJH	
MATERIAL CAST STEEL			APPD 5:1:78 AWB	(TITLE OF DWG.)
DRAFTING STANDARD AS 1100			FINISH AS MACHINED	
			SITE A3	DRG No A24681
			SCALE 1:2	SHEET 1 of 1

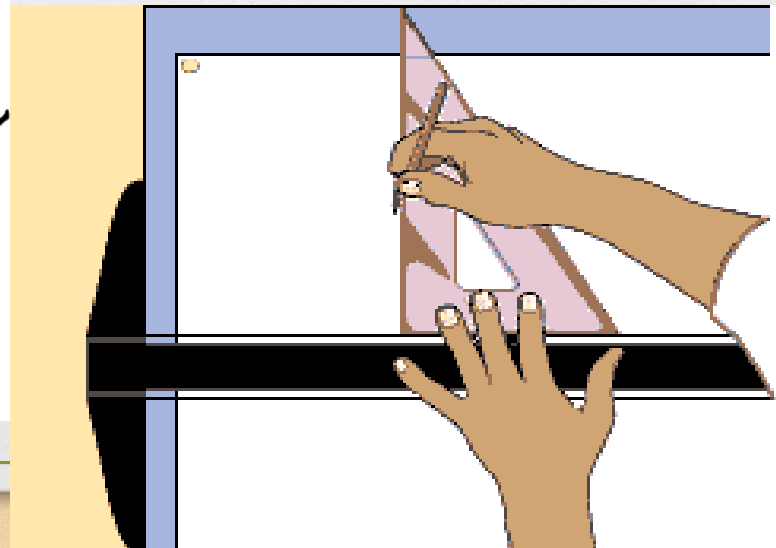
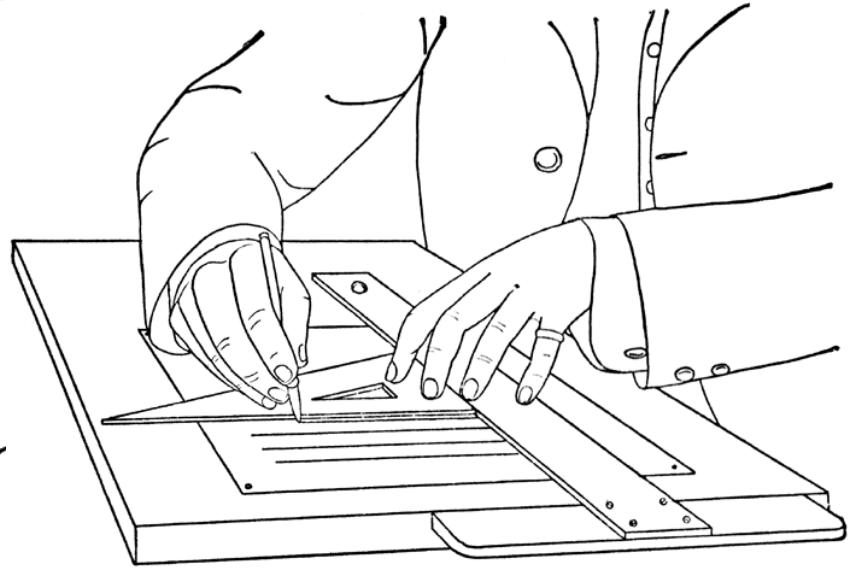
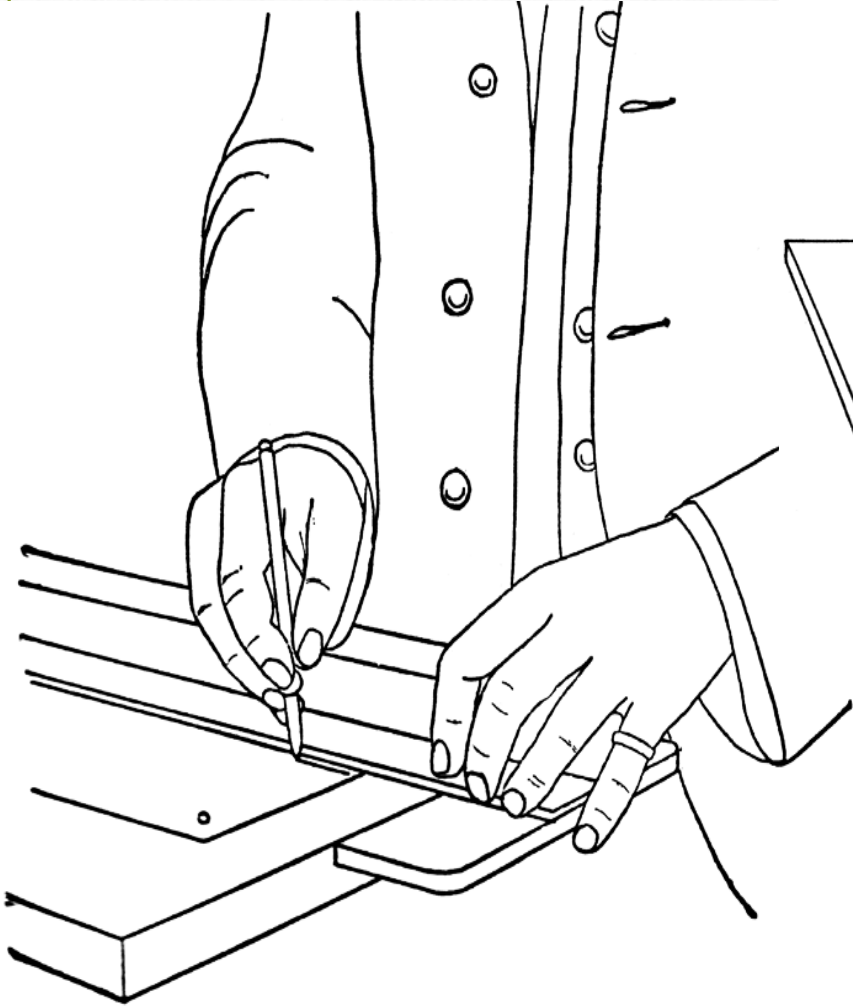
Drafting Tape



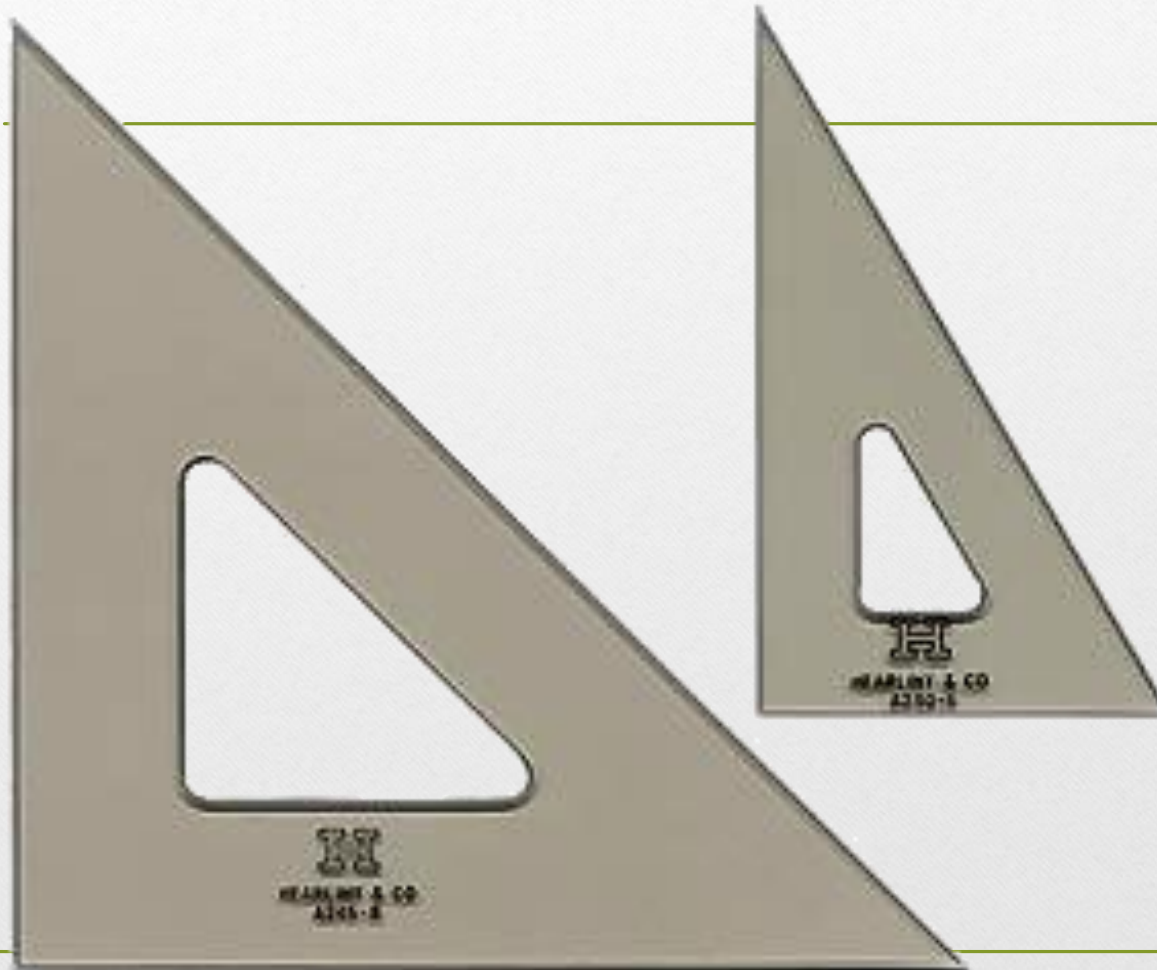
T-square

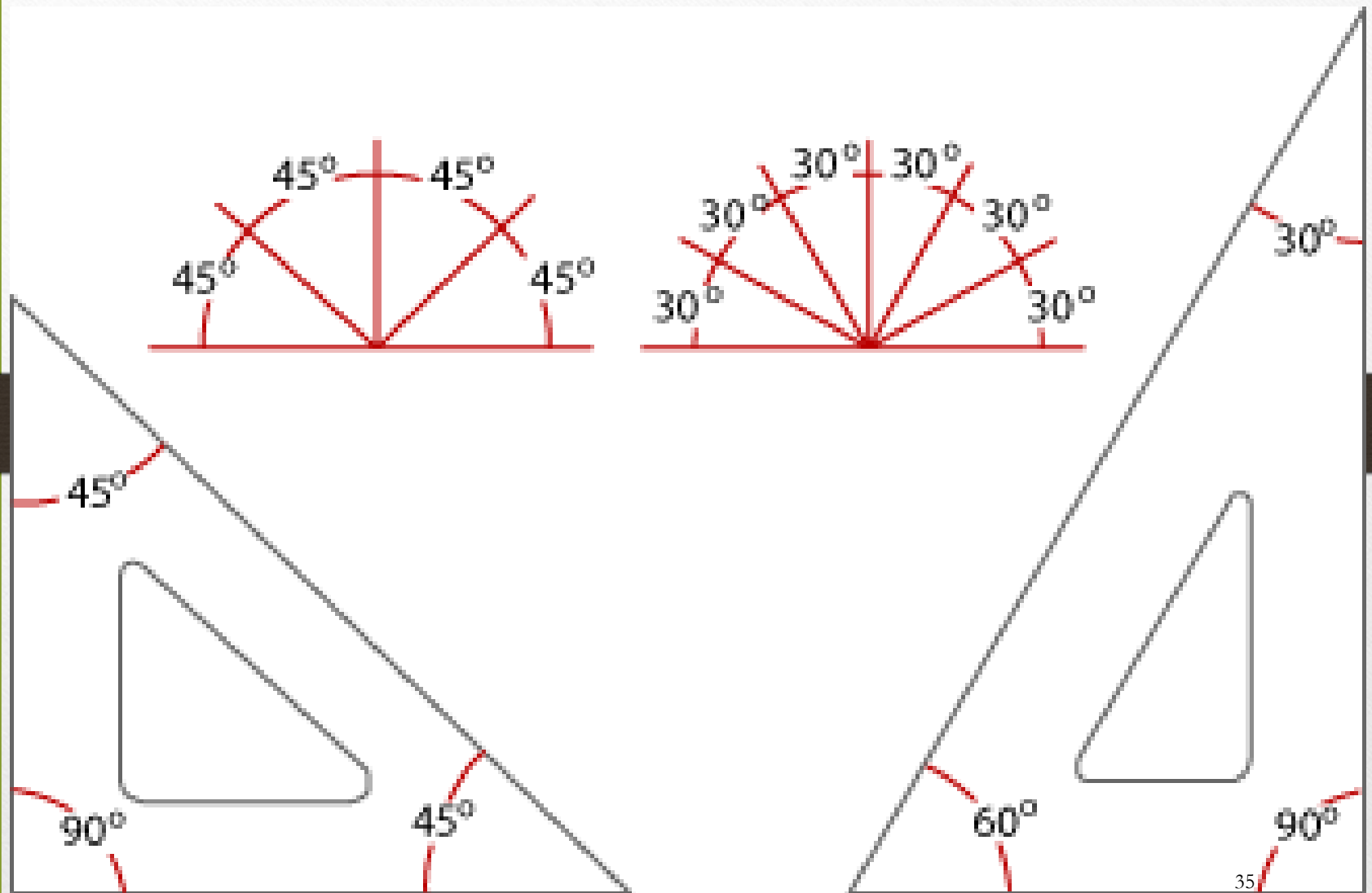


Some Techniques for Drawing



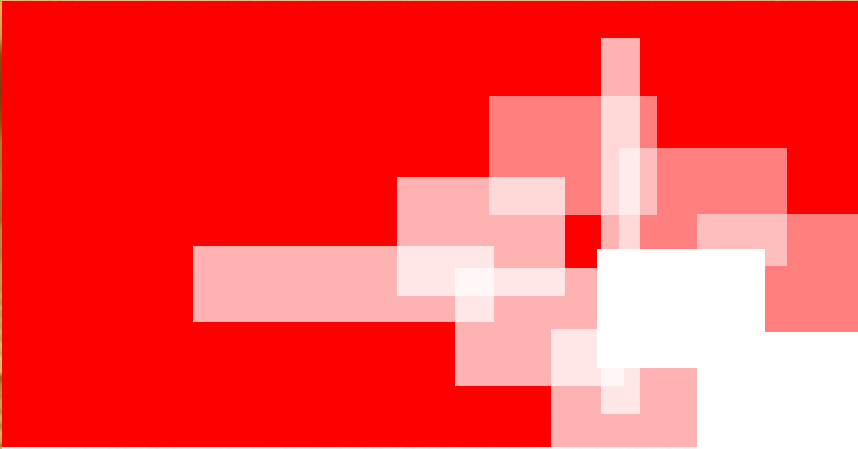
Set-squares/triangles





Compass and divider





Drawing Standards & Dimensioning

Introduction

Standards are set of rules that govern how technical drawings are represented.

Drawing standards are used so that drawings **convey the same meaning to everyone** who reads them.

Standard Code

ISO International Standards Organization

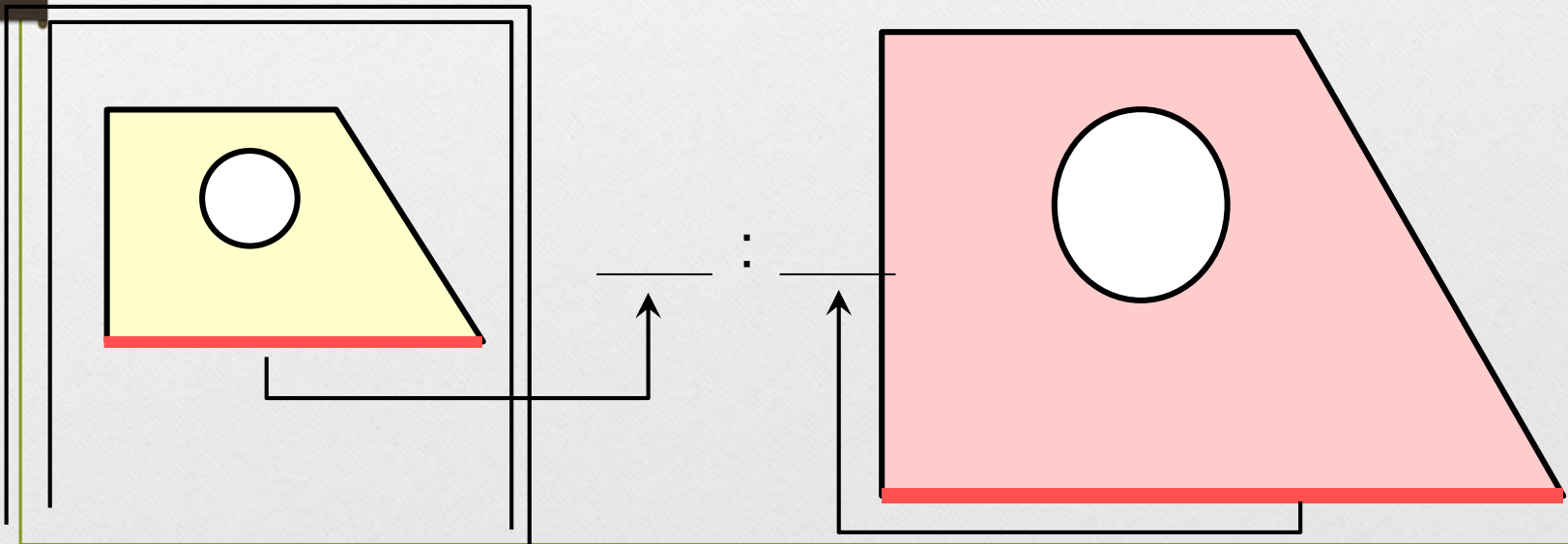
Drawing Scales

Length, size

Scale is the ratio of the linear dimension of an element of an object shown in the drawing to the real linear dimension of the same element of the object.

Size in drawing

Actual size



Meaning of Lines

Visible lines represent features that can be seen in the current view

Hidden lines represent features that can not be seen in the current view

Center line represents symmetry, path of motion, centers of circles, axis of axisymmetrical parts

Dimension and Extension lines indicate the sizes and location of features on a drawing



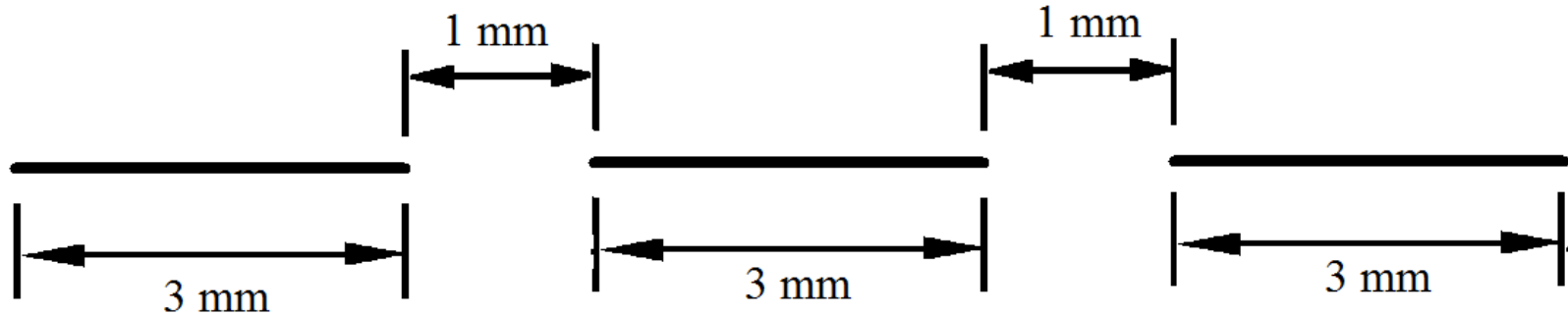
Object Line

Thickness: 100 %



Hidden Line

Thickness: 50 %



**Section Line or
Cutting Plane Line**

Thickness: 50 %



Center Line

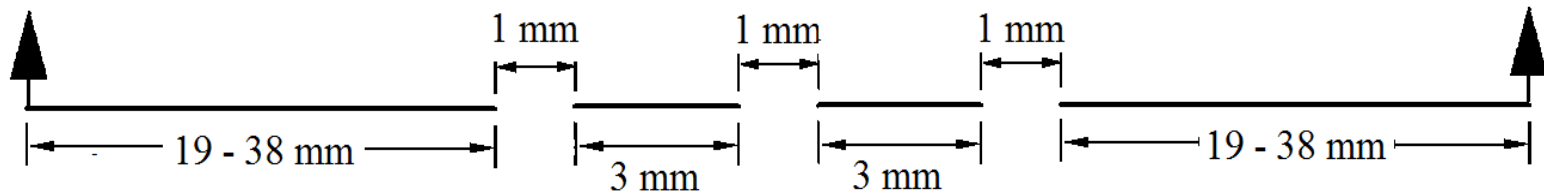
Dimension and
Extension Lines

Light
3.000

Thickness: 25 %

Section Line

Thickness: 125 %



N.B.: All Percentages are with respect to the object line

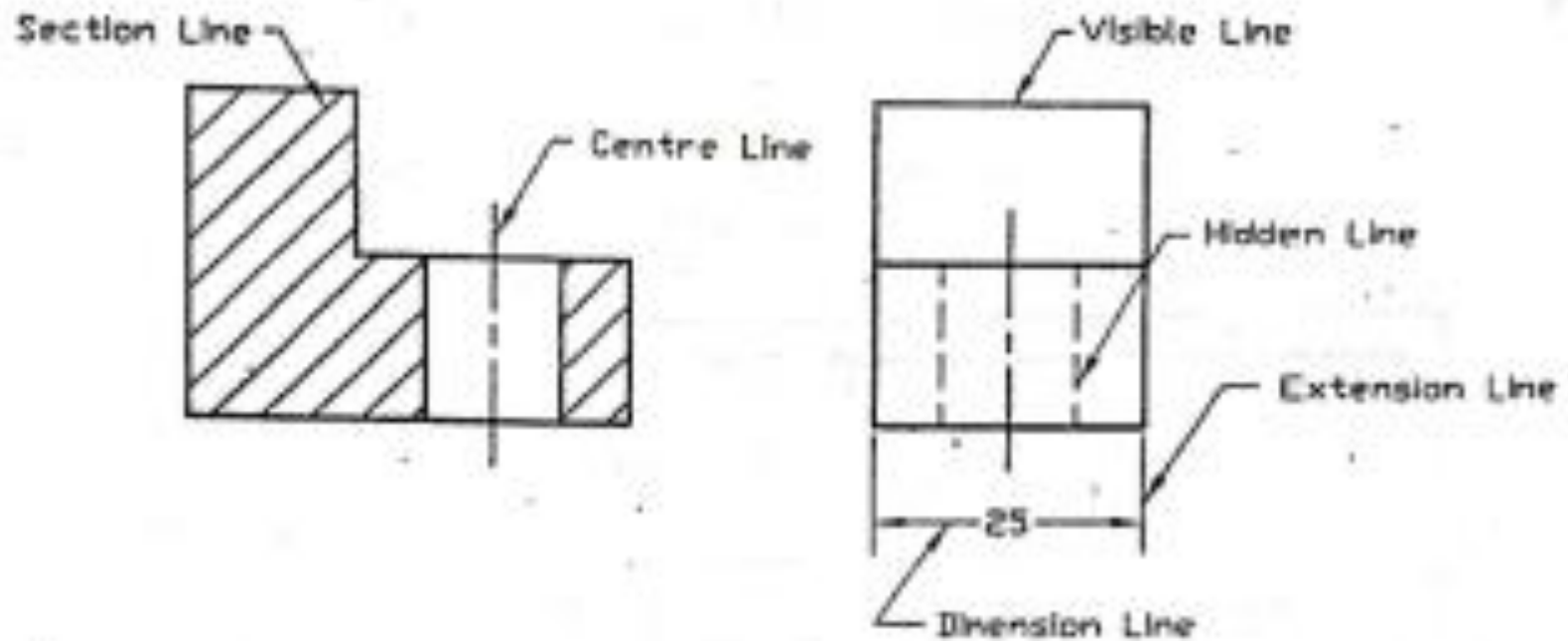


Figure 1.25: Usage of Some Common Lines•

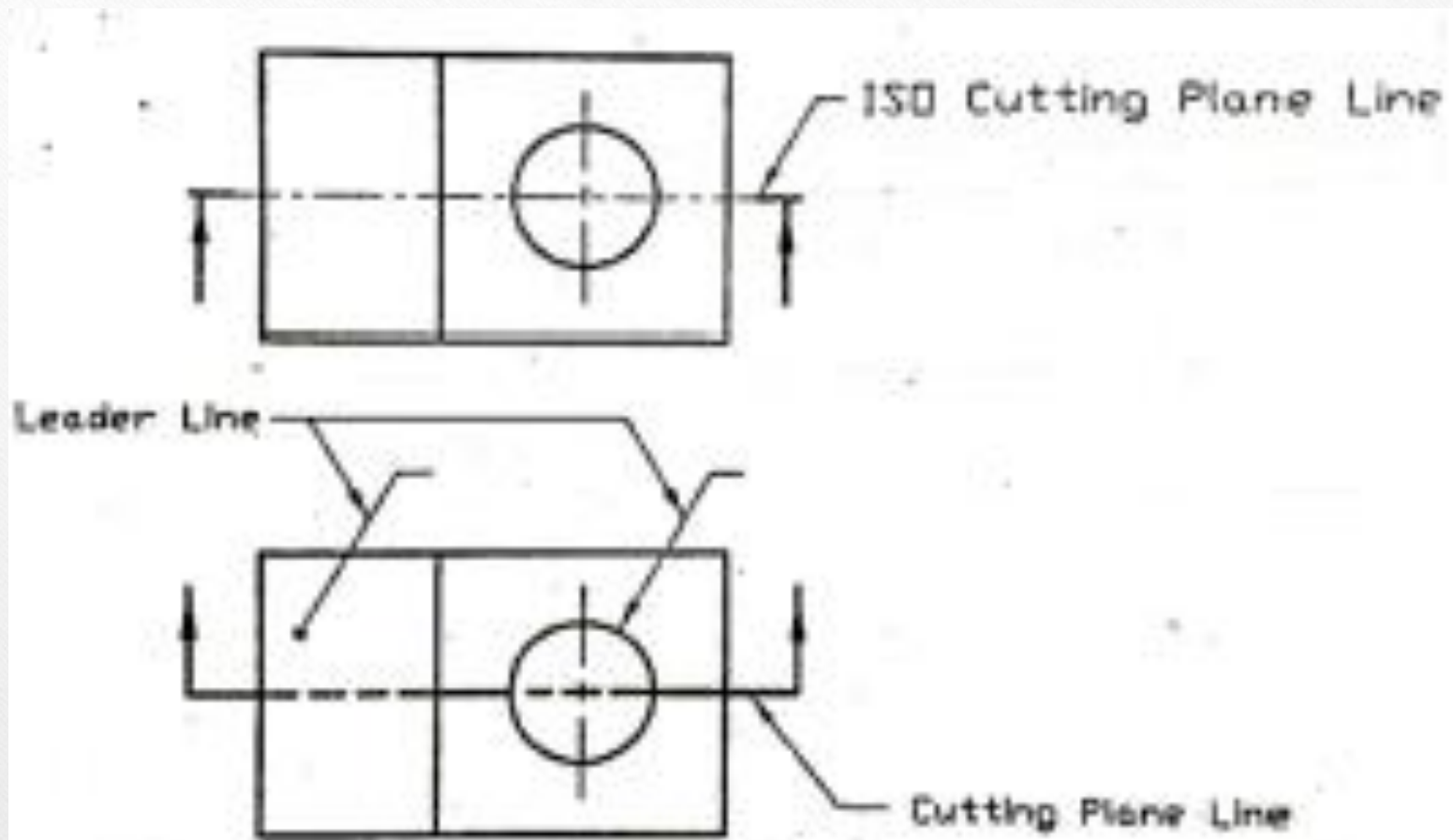


Figure 1.26: Usage of Leader and Cutting Plane Lines

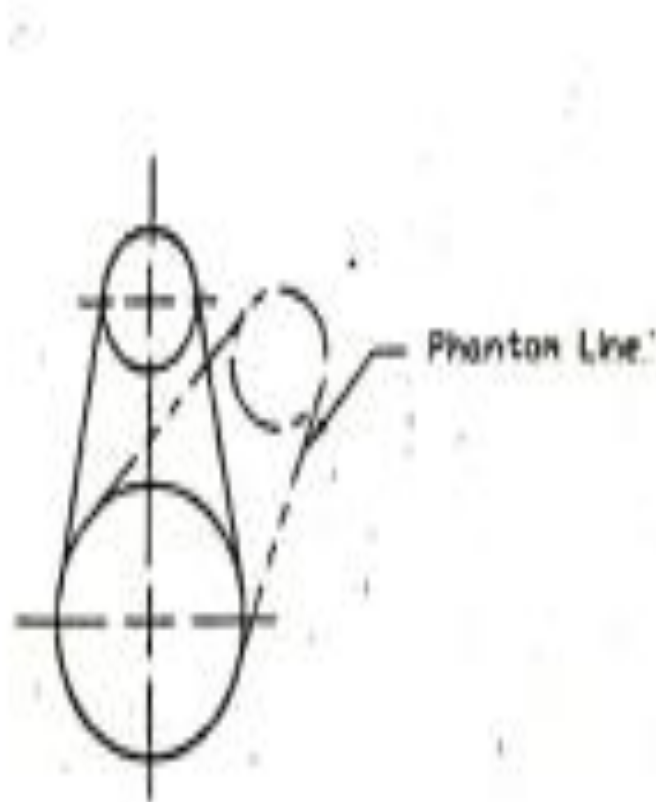


Figure 1.27: Use of Phantom Line

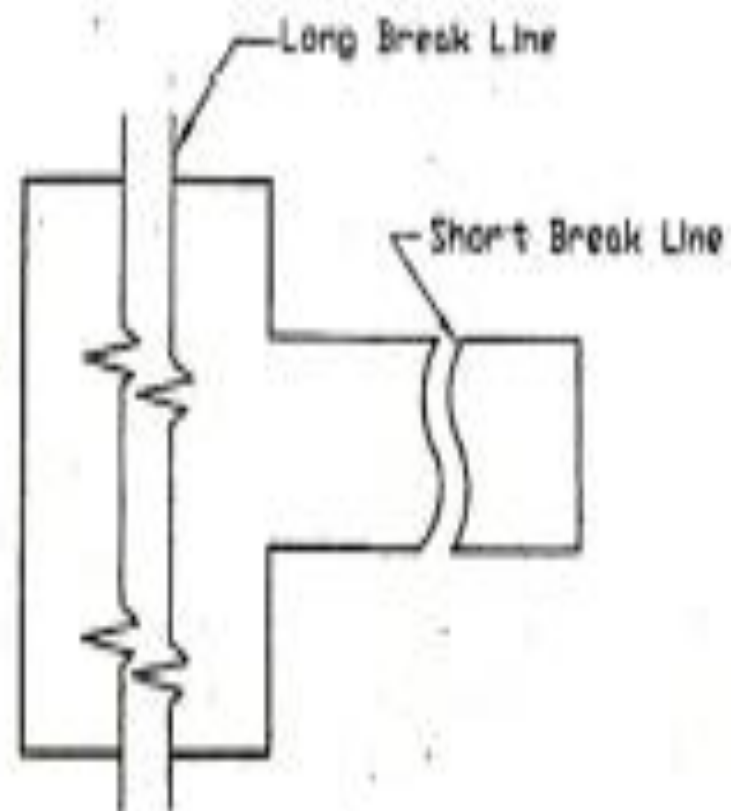
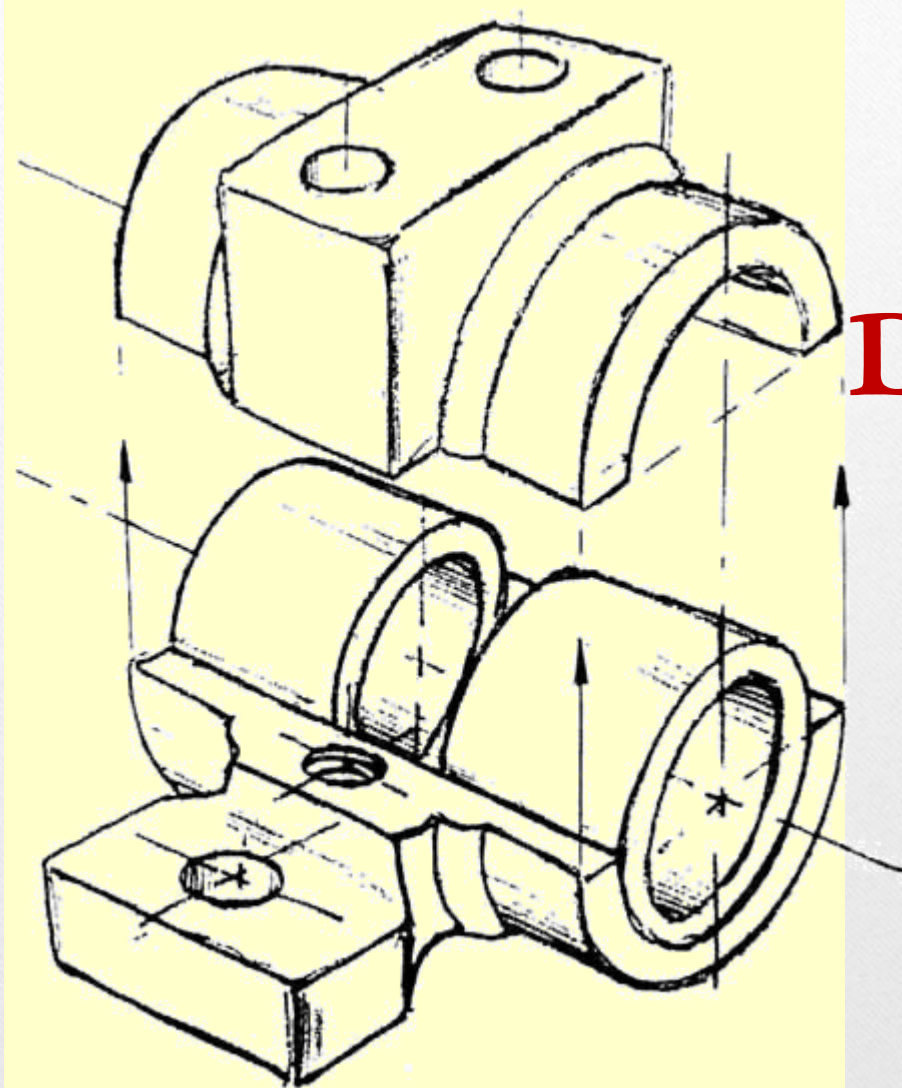
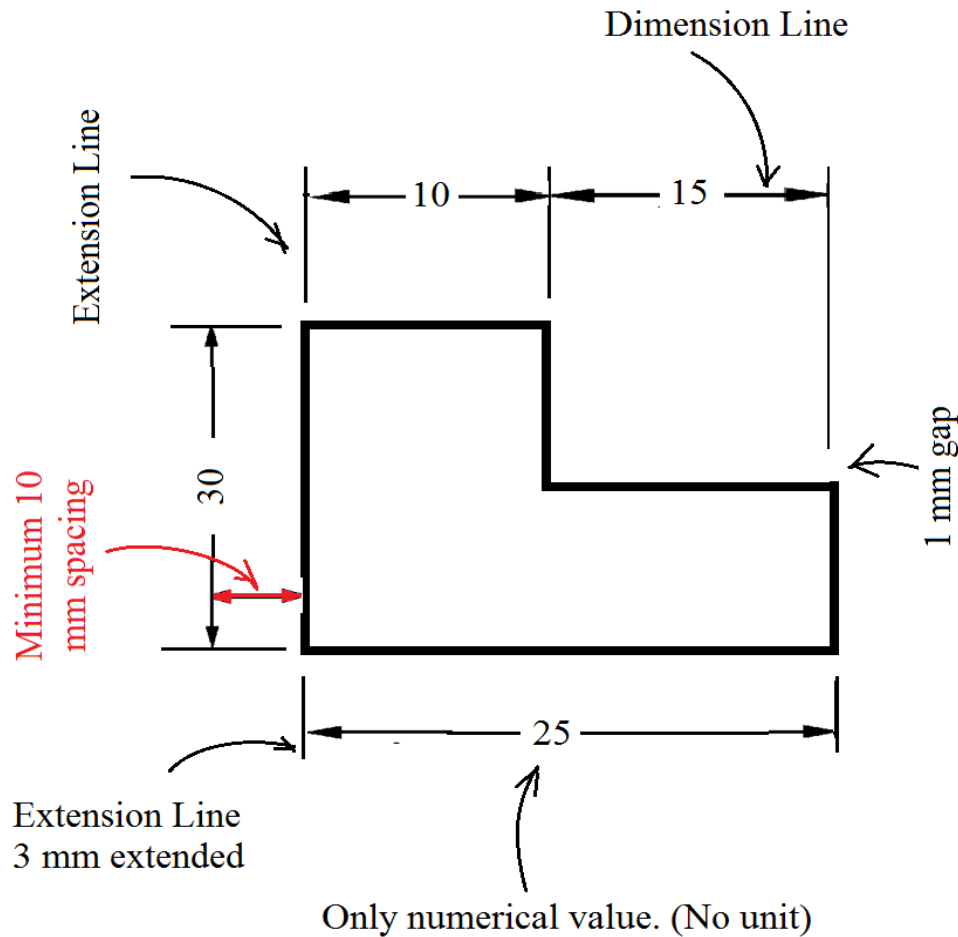


Figure 1.28: Usage of Long and Short Break Lines



Dimensioning

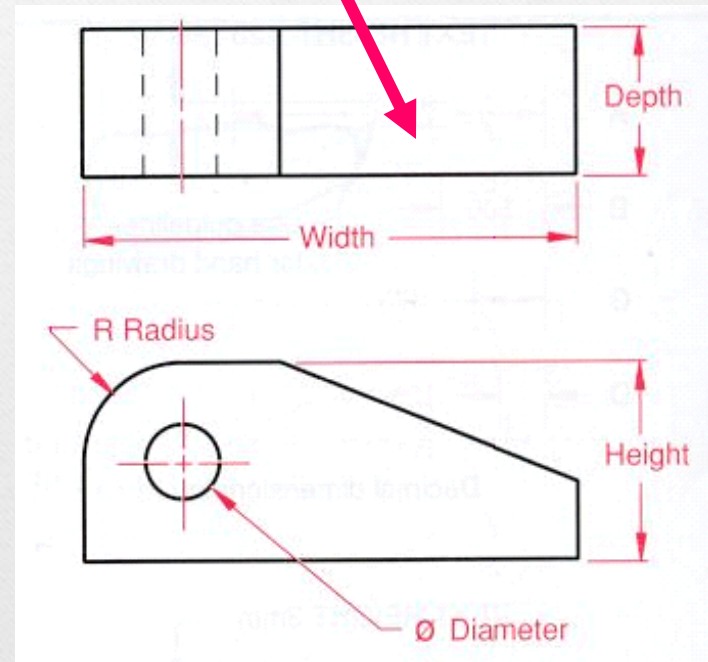
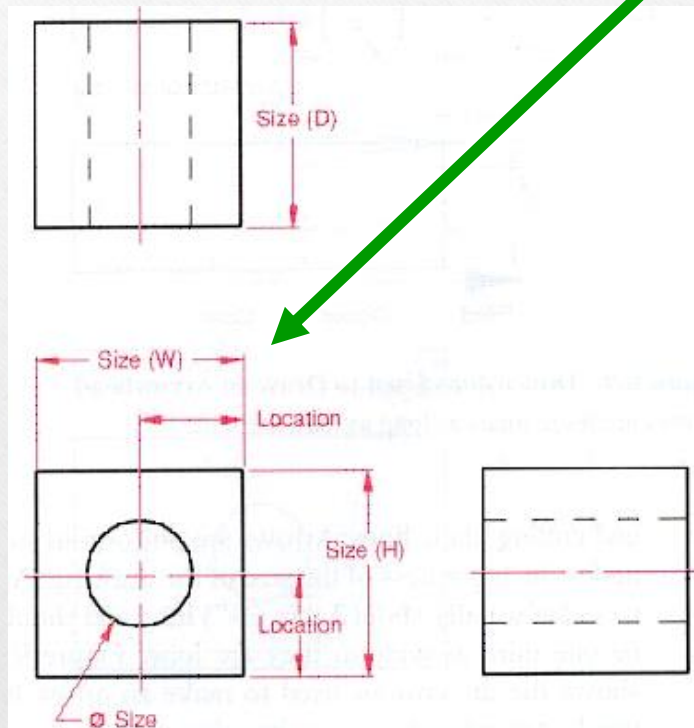
Dimensioning Guidelines



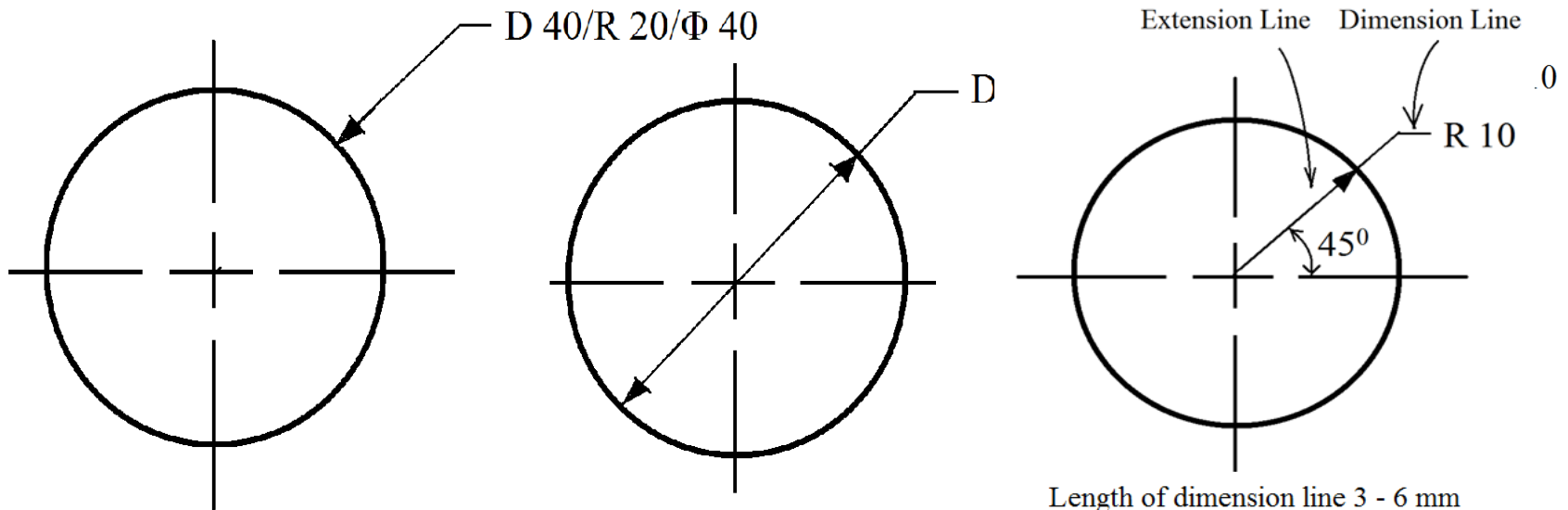
1. Dimension and extension lines must be either horizontal or vertical.
2. No dimension/extension line can cross over another dimension/extension line.
3. There must be one arrowhead at each end of dimension line.
4. The thickness of both extension and dimension line is 25%.
5. No dimensioning is allowed inside the object.

Important elements of dimensioning

Two types of dimensioning: (1) **Size and location dimensions** and (2) **Detail dimensioning**



CIRCULAR DIMENSIONING



D = Diameter;

R = Radius;

Length of dimension line 3 - 6 mm

R 10 represents Radius = 10

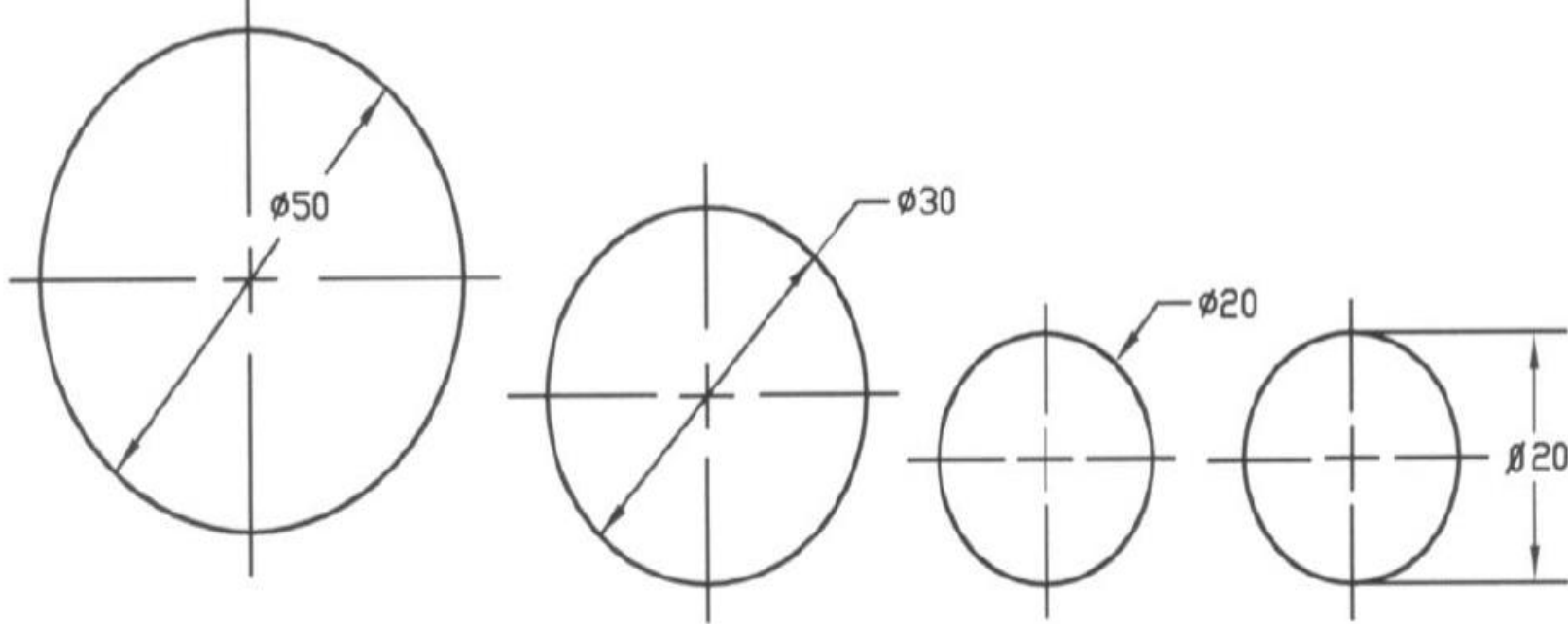
Before dimensioning Centerline must be drawn.

DRILL = Diameter;

REAM = Diameter;

The projection of the extension line in first figure
must go through the center of the circle

The arrowhead must touch the circle whose dimension is being shown.



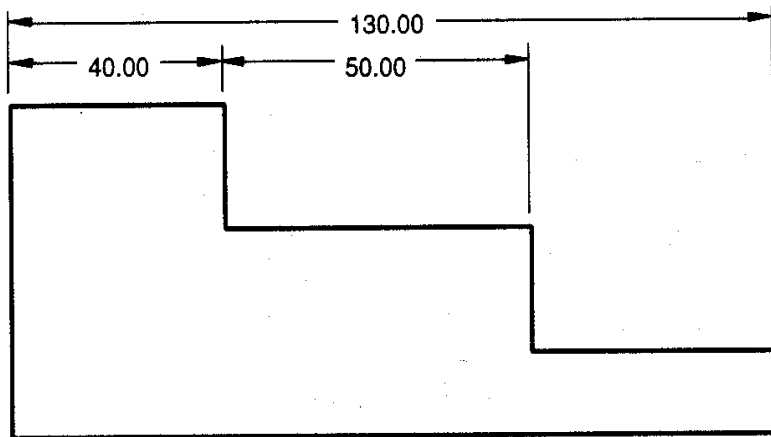
Dimensioning of Circles



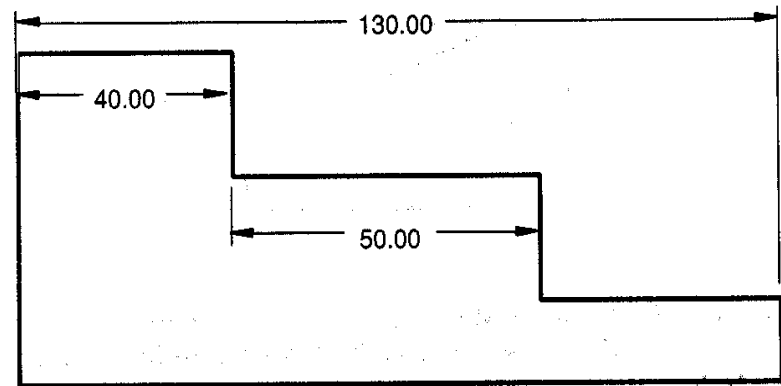
Dimensioning of Angles

Grouping Dimensions

- Dimensions should always be placed outside the part

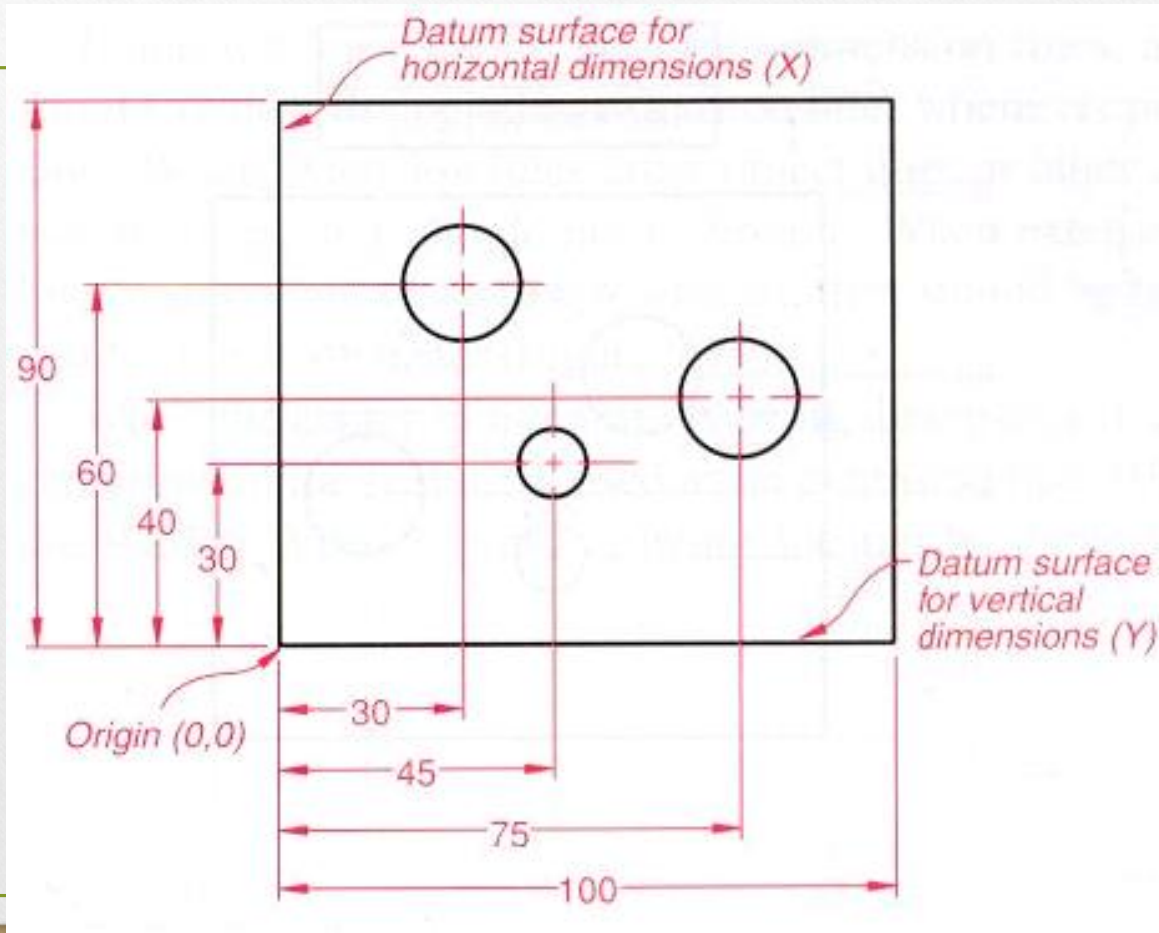


Yes

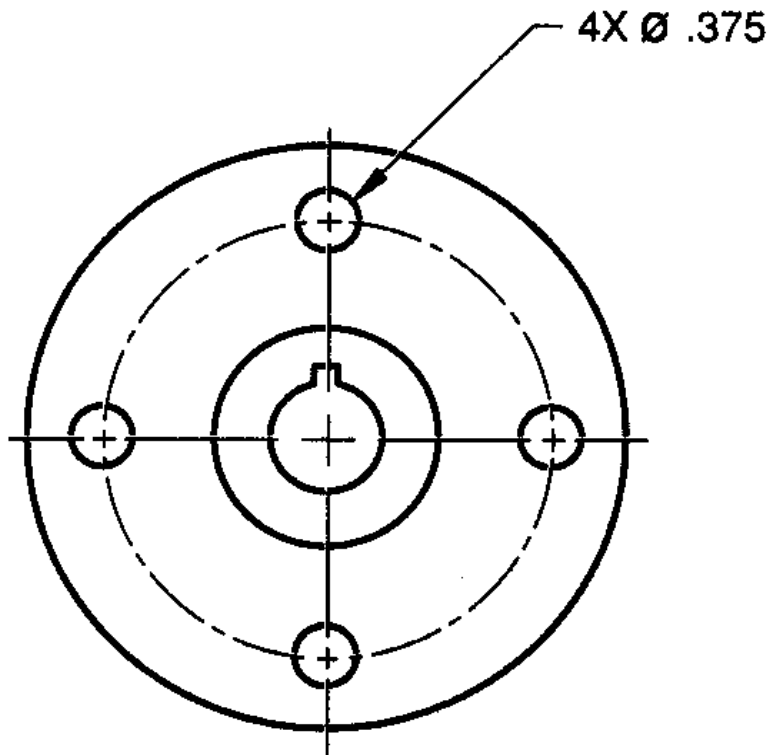


No

Where and how should we place dimensions when we have many dimensions? (cont.)

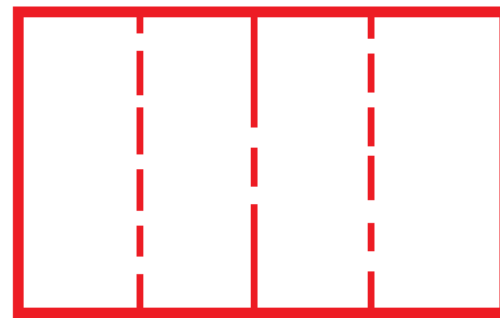
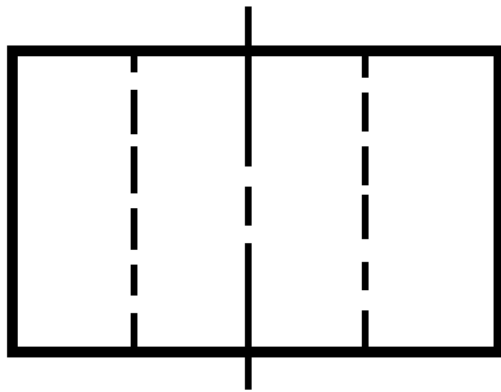
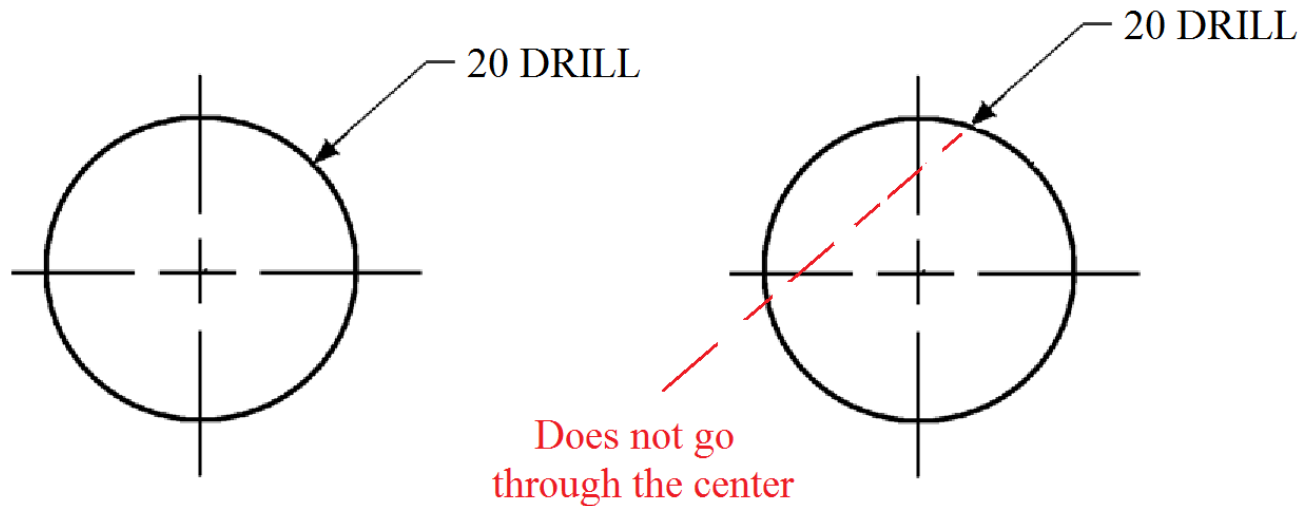


Repetitive Features



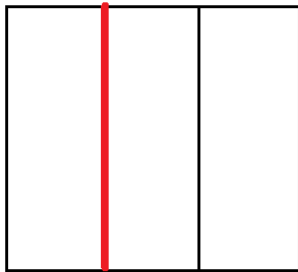
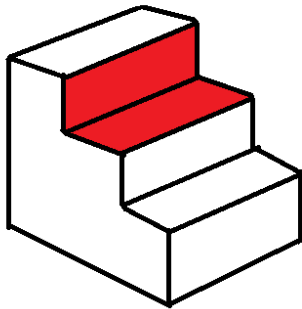
Use the Symbol 'x' to
Dimension Repetitive
Features

SOME COMMON MISTAKES

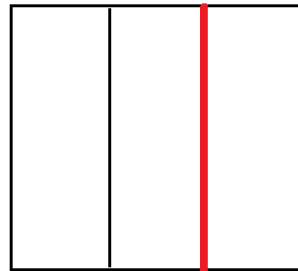
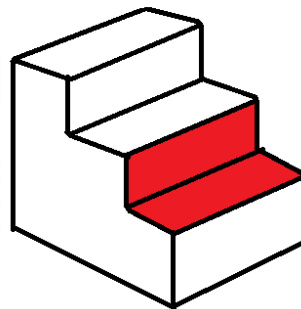


No extension for hidden line ; 3 - 6 mm extension for center line

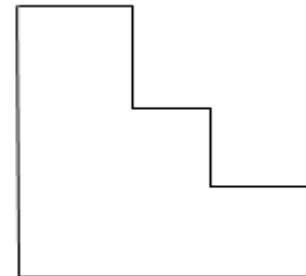
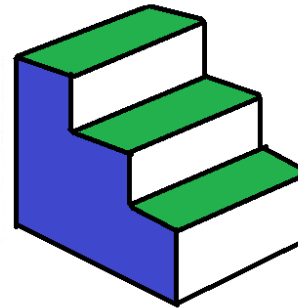
Where To Draw A Line?



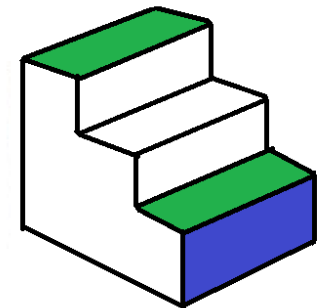
TOP VIEW



TOP VIEW



Front View



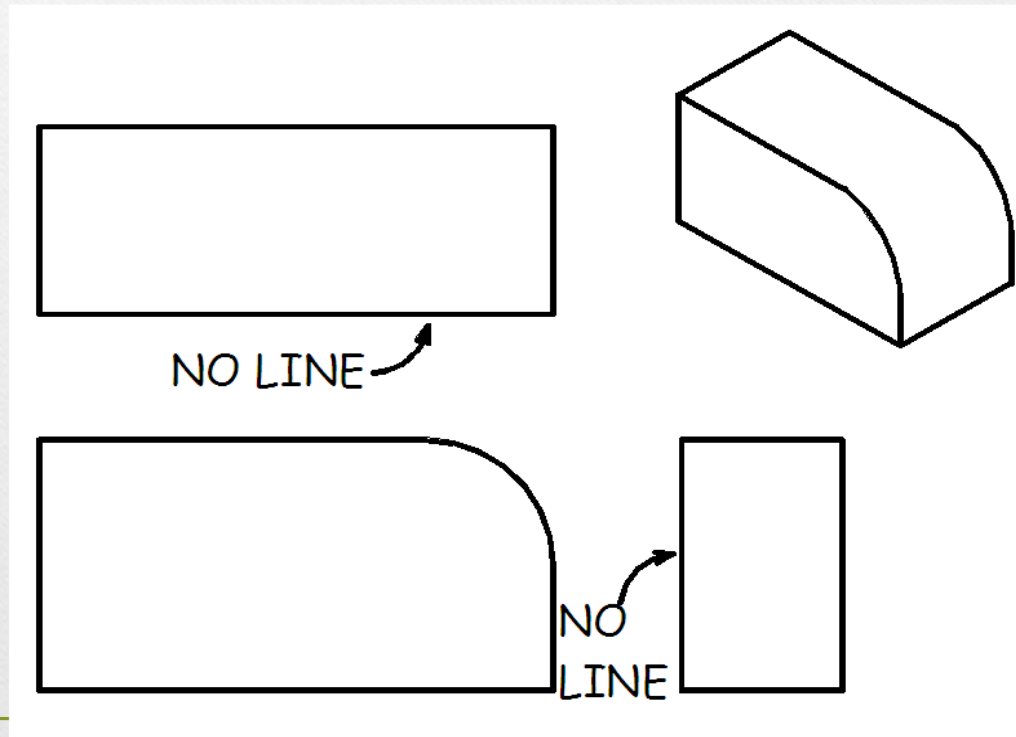
R.H.S View

Red Line is due to the intersection of two red planes shown in the isometric view.

Each segment of green line is due to the intersection of blue plane and green planes

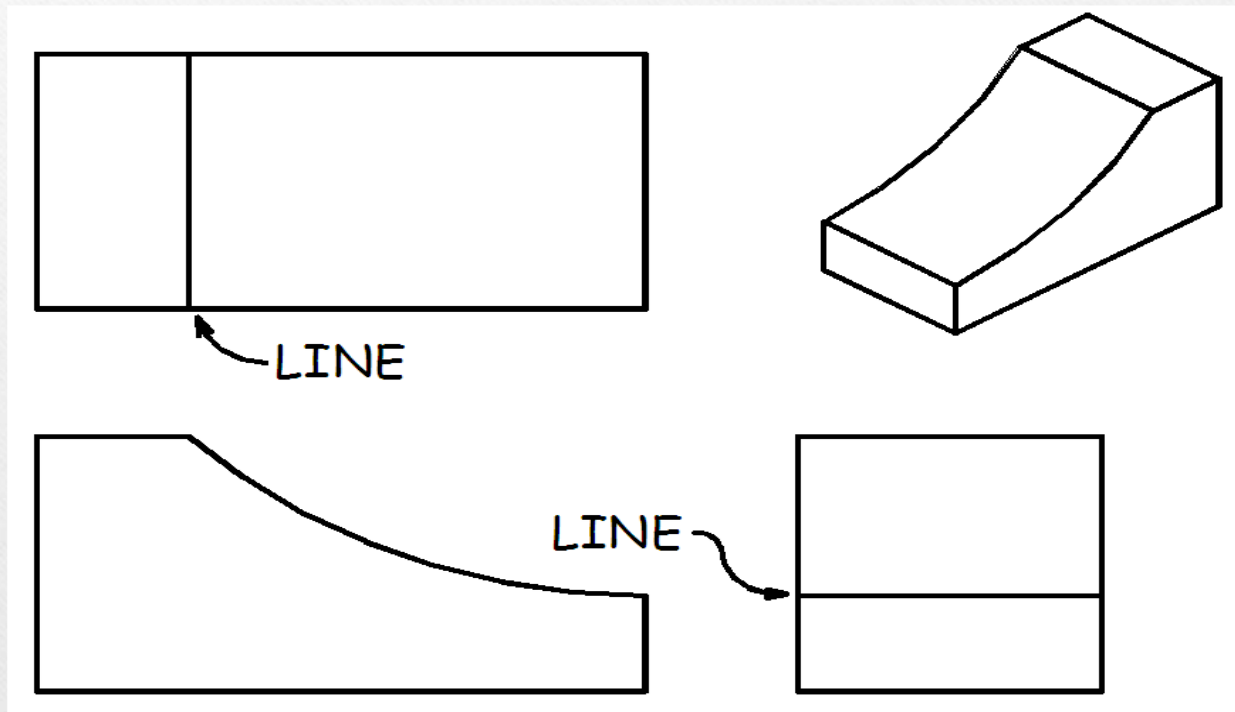
WHERE TO DRAW A LINE?

- Where a curved surface is *tangent* to a plane surface, no line should be shown where they join

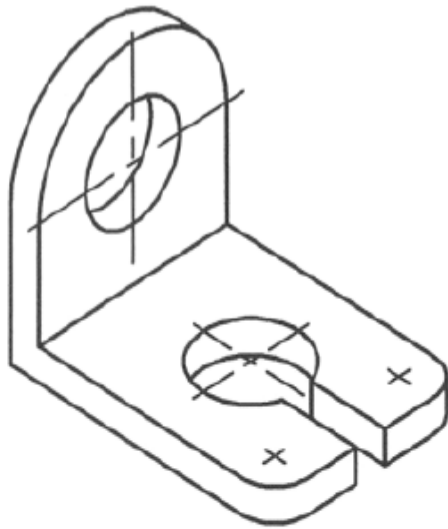


WHERE TO DRAW A LINE?

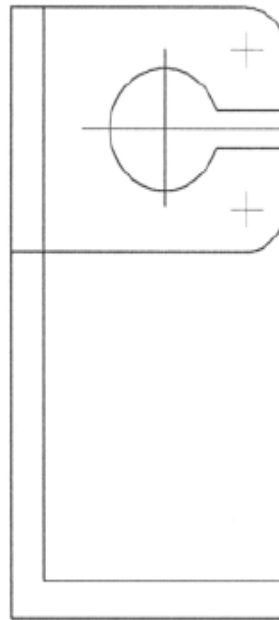
- Where a plane surface intersects a curved surface, an edge is formed



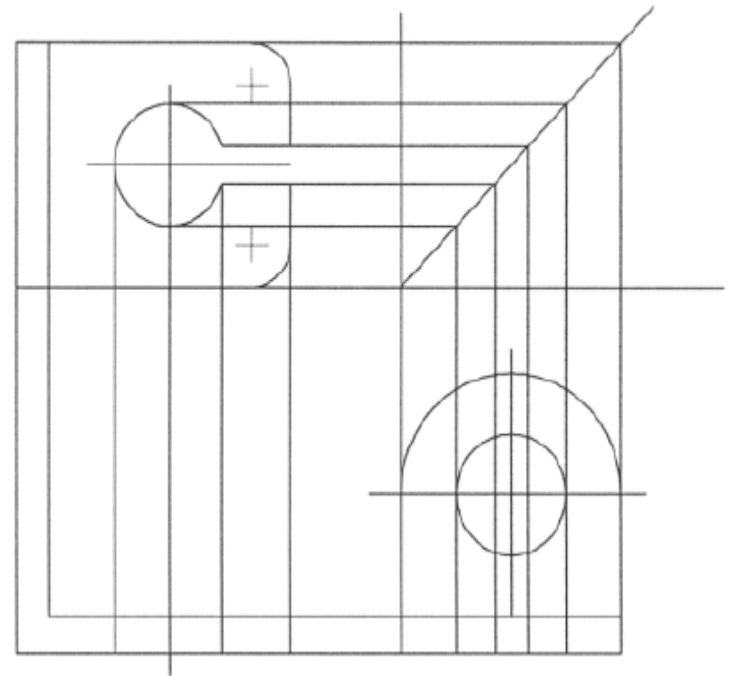
How to Start Drawing???



Step-1



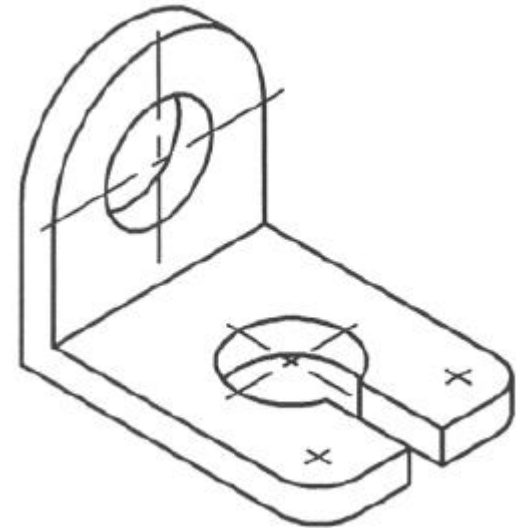
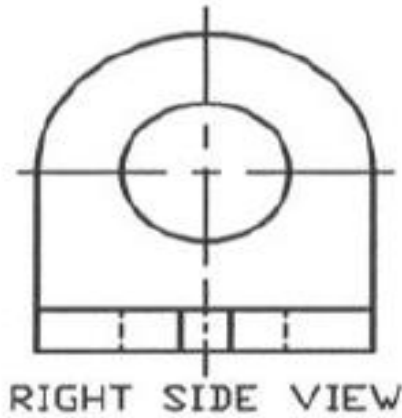
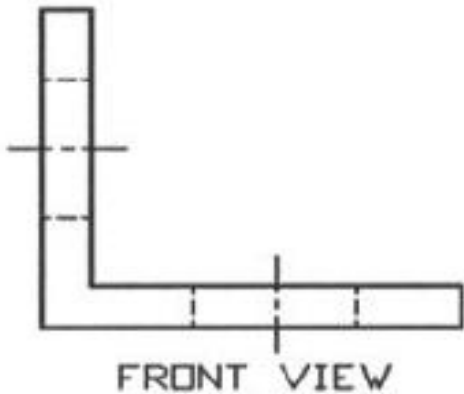
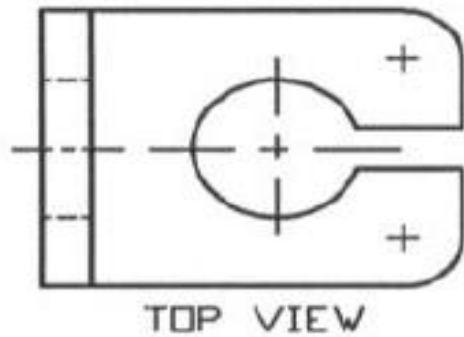
Step-2



Step-3

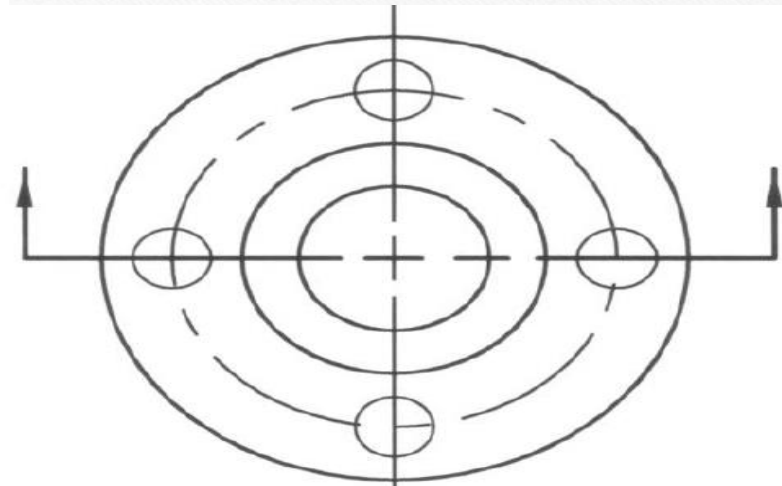
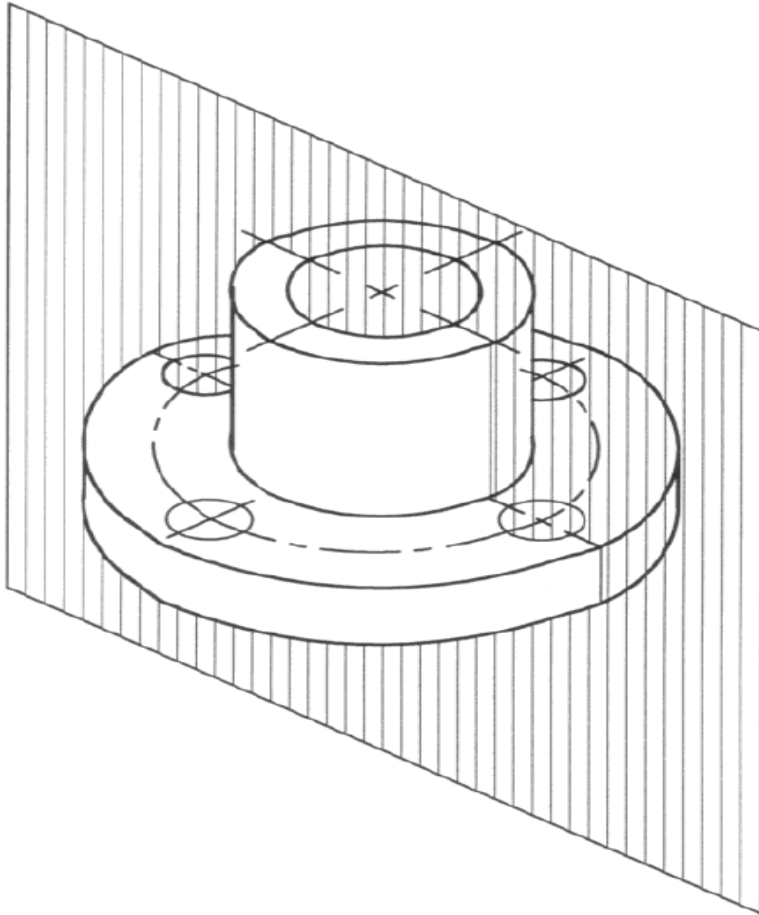
Front View

How to Start Drawing???

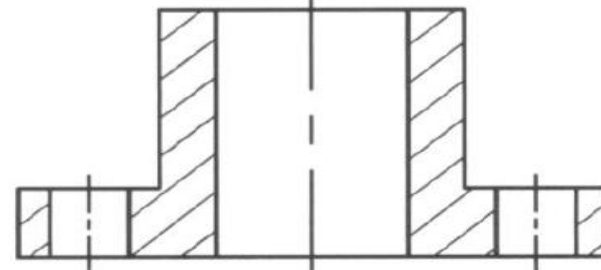


Its Done. Simple, right???

Sectional views



Top View



Sectional View

Section After Cutting and Removal of Front Portion



Cast Iron



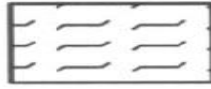
Sound
Insulation



Earth



Steel



Cork, Felt,
Leather & Fiber



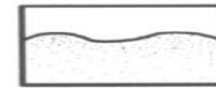
Rock



Bronze, Brass,
Copper and
Compositions



Fire Brick
and Refractory
Materials



Sand



White Metal,
Zinc, Lead,
Babbitt and
Alloys



Concrete



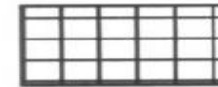
Water &
Other Liquids



Magnesium,
Aluminium



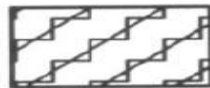
Marble, Slate,
Glass, Porcelain



Electric
Windings,
Electromagnets etc.



Rubber, Plastic,
Electrical Insulation



Thermal
Insulation



Wood

**Persist until
Succeed!**